Main Category	Category	Terms
Disease	OMIM_DISEASE	Bladder cancer, somatic
Disease	OMIM_DISEASE	Retinoblastoma
Disease	OMIM_DISEASE	Retinoblastoma, trilateral
Disease	OMIM_DISEASE	Small cell cancer of the lung, somatic
Disease	OMIM_DISEASE	Osteosarcoma, somatic
Disease	UP_KW_DISEASE	Tumor suppressor
Disease	UP_KW_DISEASE	Disease variant
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Cell cycle
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Transcription regulation
Functional Annotations	UP_KW_BIOLOGICAL_PROCESS	Host-virus interaction
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Nucleus
Functional Annotations	UP_KW_CELLULAR_COMPONENT	Cytoplasm
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Chromatin regulator
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	DNA-binding
Functional Annotations	UP_KW_MOLECULAR_FUNCTION	Repressor
Functional Annotations	UP_KW_PTM	Acetylation
Functional Annotations	UP_KW_PTM	Methylation
Functional Annotations	UP_KW_PTM	Phosphoprotein
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Acidic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic and acidic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Basic residues
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Low complexity

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	COMPBIAS:Polar residues
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Cyclin-like
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Retinoblastoma-associated protein A-box
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Retinoblastoma-associated protein B-box
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Retinoblastoma-associated protein C-terminal
Functional Annotations	UP_SEQ_FEATURE	DOMAIN:Retinoblastoma-associated protein N-terminal
Functional Annotations	UP_SEQ_FEATURE	MOTIF:Bipartite nuclear localization signal
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Abolishes monomethylation by SMYD2 and subsequent interaction with L3MBTL1.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:K->R: Does not affect the ability to be methylated by SMYD2; when associated with 873-R-R-874.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:KK->R: Does not affect the ability to be methylated by SMYD2; when associated with 873-R-R-874.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:KK->RR: Does not alter Rb localization in cycling cells, but mislocalizes to the cytoplasm during keratinocyte differentiation. Does not affect the ability to arrest cell growth. Probable loss of acetylation by PCAF.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Abolishes interaction with Pocket domain; when associated with A-821.
Functional Annotations	UP_SEQ_FEATURE	MUTAGEN:T->A: Abolishes interaction with Pocket domain; when associated with A-826.
Functional Annotations	UP_SEQ_FEATURE	REGION:Disordered
Functional Annotations	UP_SEQ_FEATURE	REGION:Domain A
Functional Annotations	UP_SEQ_FEATURE	REGION:Domain B
Functional Annotations	UP_SEQ_FEATURE	REGION:Domain C; mediates interaction with E4F1
Functional Annotations	UP_SEQ_FEATURE	REGION:Interaction with LIMD1
Functional Annotations	UP_SEQ_FEATURE	REGION:Pocket; binds T and E1A

Main Category	Category	Terms
Functional Annotations	UP_SEQ_FEATURE	REGION:Spacer
Gene Ontology	GOTERM_BP_DIRECT	G1/S transition of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	regulation of cell growth
Gene Ontology	GOTERM_BP_DIRECT	tissue homeostasis
Gene Ontology	GOTERM_BP_DIRECT	chondrocyte differentiation
Gene Ontology	GOTERM_BP_DIRECT	aortic valve morphogenesis
Gene Ontology	GOTERM_BP_DIRECT	chromatin organization
Gene Ontology	GOTERM_BP_DIRECT	chromatin remodeling
Gene Ontology	GOTERM_BP_DIRECT	regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of protein kinase activity
Gene Ontology	GOTERM_BP_DIRECT	apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	smoothened signaling pathway
Gene Ontology	GOTERM_BP_DIRECT	Ras protein signal transduction
Gene Ontology	GOTERM_BP_DIRECT	spermatogenesis
Gene Ontology	GOTERM_BP_DIRECT	regulation of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell population proliferation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of gene expression

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	glial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	neuron differentiation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell growth
Gene Ontology	GOTERM_BP_DIRECT	sister chromatid biorientation
Gene Ontology	GOTERM_BP_DIRECT	neuron projection development
Gene Ontology	GOTERM_BP_DIRECT	heterochromatin formation
Gene Ontology	GOTERM_BP_DIRECT	developmental process
Gene Ontology	GOTERM_BP_DIRECT	cellular response to insulin stimulus
Gene Ontology	GOTERM_BP_DIRECT	maintenance of mitotic sister chromatid cohesion
Gene Ontology	GOTERM_BP_DIRECT	glial cell apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	skeletal muscle cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	neuron maturation
Gene Ontology	GOTERM_BP_DIRECT	enucleate erythrocyte differentiation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of DNA-binding transcription factor activity
Gene Ontology	GOTERM_BP_DIRECT	regulation of lipid kinase activity
Gene Ontology	GOTERM_BP_DIRECT	myoblast differentiation
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of macrophage differentiation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cell cycle
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of mitotic metaphase/anaphase transition
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of smoothened signaling pathway

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of DNA-templated transcription
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription by RNA polymerase II
Gene Ontology	GOTERM_BP_DIRECT	digestive tract development
Gene Ontology	GOTERM_BP_DIRECT	cell morphogenesis involved in neuron differentiation
Gene Ontology	GOTERM_BP_DIRECT	epithelial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of epithelial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of inflammatory response
Gene Ontology	GOTERM_BP_DIRECT	striated muscle cell differentiation
Gene Ontology	GOTERM_BP_DIRECT	chromosome organization
Gene Ontology	GOTERM_BP_DIRECT	cell division
Gene Ontology	GOTERM_BP_DIRECT	neuron apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	regulation of cell cycle
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of glial cell proliferation
Gene Ontology	GOTERM_BP_DIRECT	protein localization to chromosome, centromeric region
Gene Ontology	GOTERM_BP_DIRECT	cellular response to xenobiotic stimulus
Gene Ontology	GOTERM_BP_DIRECT	hepatocyte apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of cold-induced thermogenesis
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of extracellular matrix organization
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of hepatocyte apoptotic process
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of collagen fibril organization

Main Category	Category	Terms
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of myofibroblast differentiation
Gene Ontology	GOTERM_BP_DIRECT	regulation of G1/S transition of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of G1/S transition of mitotic cell cycle
Gene Ontology	GOTERM_BP_DIRECT	positive regulation of transcription regulatory region DNA binding
Gene Ontology	GOTERM_BP_DIRECT	negative regulation of apoptotic signaling pathway
Gene Ontology	GOTERM_CC_DIRECT	chromatin
Gene Ontology	GOTERM_CC_DIRECT	nucleus
Gene Ontology	GOTERM_CC_DIRECT	nucleoplasm
Gene Ontology	GOTERM_CC_DIRECT	transcription regulator complex
Gene Ontology	GOTERM_CC_DIRECT	cytoplasm
Gene Ontology	GOTERM_CC_DIRECT	spindle
Gene Ontology	GOTERM_CC_DIRECT	cytosol
Gene Ontology	GOTERM_CC_DIRECT	cyclin/CDK positive transcription elongation factor complex
Gene Ontology	GOTERM_CC_DIRECT	SWI/SNF complex
Gene Ontology	GOTERM_CC_DIRECT	PML body
Gene Ontology	GOTERM_CC_DIRECT	Rb-E2F complex
Gene Ontology	GOTERM_CC_DIRECT	chromatin lock complex
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II transcription regulatory region sequence-specific DNA binding
Gene Ontology	GOTERM_MF_DIRECT	DNA binding
Gene Ontology	GOTERM_MF_DIRECT	transcription corepressor activity
Gene Ontology	GOTERM_MF_DIRECT	protein binding

Main Category	Category	Terms
Gene Ontology	GOTERM_MF_DIRECT	enzyme binding
Gene Ontology	GOTERM_MF_DIRECT	kinase binding
Gene Ontology	GOTERM_MF_DIRECT	ubiquitin protein ligase binding
Gene Ontology	GOTERM_MF_DIRECT	identical protein binding
Gene Ontology	GOTERM_MF_DIRECT	phosphoprotein binding
Gene Ontology	GOTERM_MF_DIRECT	molecular adaptor activity
Gene Ontology	GOTERM_MF_DIRECT	RNA polymerase II-specific DNA-binding transcription factor binding
Gene Ontology	GOTERM_MF_DIRECT	importin-alpha family protein binding
Gene Ontology	GOTERM_MF_DIRECT	disordered domain specific binding
Gene Ontology	GOTERM_MF_DIRECT	DNA-binding transcription factor binding
General Annotations	CHROMOSOME	13
General Annotations	CYTOGENETIC_LOCATION	13q14.2
General Annotations	ENTREZ_GENE_SUMMARY	The protein encoded by this gene is a negative regulator of the cell cycle and was the first tumor suppressor gene found. The encoded protein also stabilizes constitutive heterochromatin to maintain the overall chromatin structure. The active, hypophosphorylated form of the protein binds transcription factor E2F1. Defects in this gene are a cause of childhood cancer retinoblastoma (RB), bladder cancer, and osteogenic sarcoma. [provided by RefSeq, Jul 2008]
General Annotations	UP_COMMENT_DISEASE	Bladder cancer (BLC) [MIM:109800]: A malignancy originating in tissues of the urinary bladder. It often presents with multiple tumors appearing at different times and at different sites in the bladder. Most bladder cancers are transitional cell carcinomas that begin in cells that normally make up the inner lining of the bladder. Other types of bladder cancer include squamous cell carcinoma (cancer that begins in thin, flat cells) and adenocarcinoma (cancer that begins in cells that make and release mucus and other fluids). Bladder cancer is a complex disorder with both genetic and environmental influences. Note=Disease susceptibility is associated with variants affecting the gene represented in this entry.
General	UP_COMMENT_DISEASE	Childhood cancer retinoblastoma (RB) [MIM:180200]:

Main Category	Category	Terms
Annotations		Congenital malignant tumor that arises from the nuclear layers of the retina. It occurs in about 1:20'000 live births and represents about 2% of childhood malignancies. It is bilateral in about 30% of cases. Although most RB appear sporadically, about 20% are transmitted as an autosomal dominant trait with incomplete penetrance. The diagnosis is usually made before the age of 2 years when strabismus or a gray to yellow reflex from pupil ('cat eye') is investigated. {ECO:0000269|PubMed:10671068, ECO:0000269|PubMed:11524739, ECO:0000269|PubMed:1352883, ECO:0000269|PubMed:2594029, ECO:0000269|PubMed:7704558, ECO:0000269|PubMed:7795591, ECO:0000269|PubMed:7927327, ECO:0000269|PubMed:8118465, ECO:0000269|PubMed:8346255, ECO:0000269|PubMed:8605116, ECO:0000269|PubMed:8776589, ECO:0000269|PubMed:9140452, ECO:0000269|PubMed:9311732, ECO:0000269|PubMed:9973307}. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DISEASE	Osteogenic sarcoma (OSRC) [MIM:259500]: A sarcoma originating in bone-forming cells, affecting the ends of long bones. Note=The disease is caused by variants affecting the gene represented in this entry.
General Annotations	UP_COMMENT_DOMAIN	The Pocket domain binds to the threonine-phosphorylated domain C, thereby preventing interaction with heterodimeric E2F/DP transcription factor complexes.
General Annotations	UP_COMMENT_FUNCTION	(Microbial infection) In case of viral infections, interactions with SV40 large T antigen, HPV E7 protein or adenovirus E1A protein induce the disassembly of RB1-E2F1 complex thereby disrupting RB1's activity. {ECO:0000269|PubMed:1316611, ECO:0000269|PubMed:17974914, ECO:0000269|PubMed:18701596, ECO:0000269|PubMed:2839300, ECO:0000269|PubMed:8892909}.
General Annotations	UP_COMMENT_FUNCTION	Tumor suppressor that is a key regulator of the G1/S transition of the cell cycle (PubMed:10499802). The hypophosphorylated form binds transcription regulators of the E2F family, preventing transcription of E2F-responsive genes (PubMed:10499802). Both physically blocks E2Fs transactivating domain and recruits chromatin- modifying enzymes that actively repress transcription (PubMed:10499802). Cyclin and CDK-dependent phosphorylation of RB1 induces its dissociation from E2Fs, thereby activating transcription of E2F responsive genes and triggering entry into S phase (PubMed:10499802). RB1 also promotes the G0-G1 transition upon phosphorylation and activation by CDK3/cyclin-C (PubMed:15084261). Directly involved in heterochromatin formation by maintaining overall chromatin structure and, in particular, that of constitutive heterochromatin by stabilizing histone methylation. Recruits and targets histone methyltransferases SUV39H1, KMT5B and KMT5C, leading to epigenetic transcriptional repression. Controls histone H4 'Lys-20' trimethylation. Inhibits the

Main Category	Category	Terms
		intrinsic kinase activity of TAF1. Mediates transcriptional repression by SMARCA4/BRG1 by recruiting a histone deacetylase (HDAC) complex to the c-FOS promoter. In resting neurons, transcription of the c-FOS promoter is inhibited by BRG1- dependent recruitment of a phospho-RB1-HDAC1 repressor complex. Upon calcium influx, RB1 is dephosphorylated by calcineurin, which leads to release of the repressor complex (By similarity). {ECO:0000250|UniProtKB:P13405, ECO:0000250|UniProtKB:P33568, ECO:0000269|PubMed:10499802, ECO:0000269|PubMed:15084261}.
General Annotations	UP_COMMENT_INTERACTION	P06400; Q9UKV8: AGO2; NbExp=3; IntAct=EBI-491274, EBI-528269; P06400; P24863: CCNC; NbExp=4; IntAct=EBI-491274, EBI-395261; P06400; P24941: CDK2; NbExp=3; IntAct=EBI-491274, EBI-375096; P06400; O14757: CHEK1; NbExp=3; IntAct=EBI-491274, EBI-974488; P06400; O96017: CHEK2; NbExp=3; IntAct=EBI-491274, EBI-1180783; P06400; Q13574-2: DGKZ; NbExp=6; IntAct=EBI-491274, EBI-715527; P06400; Q13627: DYRK1A; NbExp=4; IntAct=EBI-491274, EBI-1053596; P06400; Q9Y463: DYRK1B; NbExp=4; IntAct=EBI-491274, EBI-634187; P06400; Q01094: E2F1; NbExp=29; IntAct=EBI-491274, EBI-448924; P06400; Q14209: E2F2; NbExp=5; IntAct=EBI-491274, EBI-718476; P06400; O00716: E2F3; NbExp=5; IntAct=EBI-491274, EBI-765551; P06400; Q16254: E2F4; NbExp=6; IntAct=EBI-491274, EBI-448943; P06400; Q15329: E2F5; NbExp=2; IntAct=EBI-491274, EBI-1389773; P06400; O43524: FOXO3; NbExp=2; IntAct=EBI-491274, EBI-1644164; P06400; P42685: FRK; NbExp=3; IntAct=EBI-491274, EBI-1383583; P06400; P15976: GATA1; NbExp=2; IntAct=EBI-491274, EBI-3909284; P06400; P62993: GRB2; NbExp=4; IntAct=EBI-491274, EBI-401755; P06400; O60381: HBP1; NbExp=2; IntAct=EBI-491274, EBI-954175; P06400; Q13547: HDAC1; NbExp=16; IntAct=EBI-491274, EBI-301834; P06400; Q14653: IRF3; NbExp=2; IntAct=EBI-491274, EBI-2650369; P06400; P29375: KDM5A; NbExp=2; IntAct=EBI-491274, EBI-1560836; P06400; Q16539: MAPK14; NbExp=4; IntAct=EBI-491274, EBI-73946; P06400; Q00987: MDM2; NbExp=5; IntAct=EBI-491274, EBI-389668; P06400; O15151: MDM4; NbExp=4; IntAct=EBI-491274, EBI-398437; P06400; Q9Y676: MRPS18B; NbExp=2; IntAct=EBI-491274, EBI-750085; P06400; Q14686: NCOA6; NbExp=3; IntAct=EBI-491274, EBI-78670; P06400; P07197: NEFM; NbExp=2; IntAct=EBI-491274, EBI-1105035; P06400; Q9UQ80: PA2G4; NbExp=4; IntAct=EBI-491274, EBI-924893; P06400; P62136: PPP1CA; NbExp=2; IntAct=EBI-491274, EBI-357253; P06400; P55345: PRMT2; NbExp=3; IntAct=EBI-491274, EBI-78458; P06400; Q00577: PURA; NbExp=6; IntAct=EBI-491274, EBI-1045860; P06400; P04049: RAF1; NbExp=3; IntAct=EBI-491274, EBI-365996; P06400; P06400: RB1; NbExp=3; IntAct=EBI-491274, EBI-491274; P06400; Q99708: RBBP8; NbExp=2; IntAct=EBI-491274, EBI-745715; P06400; O75150: RNF40; NbExp=3; IntAct=EBI-491274, EBI-744408; P06400; Q8WTS6: SETD7; NbExp=4; IntAct=EBI-491274,

Main Category	Category	Terms
		EBI-1268586; P06400; Q96PU4: UHRF2; NbExp=4; IntAct=EBI-491274, EBI-625304; P06400; Q93009: USP7; NbExp=8; IntAct=EBI-491274, EBI-302474; P06400; P30285: Cdk4; Xeno; NbExp=2; IntAct=EBI-491274, EBI-847225; P06400; Q155P7: Cenpf; Xeno; NbExp=4; IntAct=EBI-491274, EBI-2211248; P06400; A0MPS7: E7; Xeno; NbExp=2; IntAct=EBI-491274, EBI-7014709; P06400; P03129: E7; Xeno; NbExp=23; IntAct=EBI-491274, EBI-866453; P06400; P04020: E7; Xeno; NbExp=3; IntAct=EBI-491274, EBI-7005254; P06400; P06464: E7; Xeno; NbExp=3; IntAct=EBI-491274, EBI-6944797; P06400; P06465: E7; Xeno; NbExp=3; IntAct=EBI-491274, EBI-963841; P06400; P06788: E7; Xeno; NbExp=4; IntAct=EBI-491274, EBI-1776887; P06400; P52927: Hmga2; Xeno; NbExp=5; IntAct=EBI-491274, EBI-912574; P06400; Q9R002: Ifi202; Xeno; NbExp=5; IntAct=EBI-491274, EBI-3043899; P06400; Q8VSP9: ospF; Xeno; NbExp=2; IntAct=EBI-491274, EBI-6506625; P06400; Q9Z2X2: Psmd10; Xeno; NbExp=3; IntAct=EBI-491274, EBI-8377084; P06400; PRO_0000037321 [P0C6X7]: rep; Xeno; NbExp=3; IntAct=EBI-491274, EBI-25487301; P06400; Q923E4: Sirt1; Xeno; NbExp=4; IntAct=EBI-491274, EBI-1802585; P06400; Q3TKT4: Smarca4; Xeno; NbExp=4; IntAct=EBI-491274, EBI-1210244; P06400; P03070; Xeno; NbExp=6; IntAct=EBI-491274, EBI-617698; P06400; P03254; Xeno; NbExp=3; IntAct=EBI-491274, EBI-8599077; P06400; P03255; Xeno; NbExp=10; IntAct=EBI-491274, EBI-2603114; P06400; P03255-1; Xeno; NbExp=2; IntAct=EBI-491274, EBI-6692439; P06400; P03255-2; Xeno; NbExp=3; IntAct=EBI-491274, EBI-6859460; P06400; PRO_0000041225 [Q86500]; Xeno; NbExp=3; IntAct=EBI-491274, EBI-2568719;
General Annotations	UP_COMMENT_PTM	Acetylated during keratinocyte differentiation. Acetylation at Lys-873 and Lys-874 regulates subcellular localization. Can be deacetylated by SIRT1. {ECO:0000269|PubMed:20940255}.
General Annotations	UP_COMMENT_PTM	N-terminus is methylated by METTL11A/NTM1 (By similarity). Monomethylation at Lys-810 by SMYD2 enhances phosphorylation at Ser-807 and Ser-811, and promotes cell cycle progression. Monomethylation at Lys-860 by SMYD2 promotes interaction with L3MBTL1. {ECO:0000250, ECO:0000269|PubMed:15084261, ECO:0000269|PubMed:15809340, ECO:0000269|PubMed:16360038, ECO:0000269|PubMed:17380128, ECO:0000269|PubMed:1756735, ECO:0000269|PubMed:20870719, ECO:0000269|PubMed:22787429}.
General Annotations	UP_COMMENT_PTM	Phosphorylated by CDK6 and CDK4, and subsequently by CDK2 at Ser- 567 in G1, thereby releasing E2F1 which is then able to activate cell growth. Dephosphorylated at the late M phase. SV40 large T antigen, HPV E7 and adenovirus E1A bind to the underphosphorylated, active form of pRb. Phosphorylation at Thr-821 and Thr-826 promotes interaction between the C-terminal domain C and the Pocket domain, and thereby inhibits interactions with heterodimeric E2F/DP transcription

Main Category	Category	Terms
		factor complexes. Dephosphorylated at Ser-795 by calcineruin upon calcium stimulation. CDK3/cyclin-C-mediated phosphorylation at Ser-807 and Ser- 811 is required for G0-G1 transition. Phosphorylated by CDK1 and CDK2 upon TGFB1-mediated apoptosis. {ECO:0000269|PubMed:10499802, ECO:0000269|PubMed:8114739}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the retinoblastoma protein (RB) family. {ECO:0000256|ARBA:ARBA00009475}.
General Annotations	UP_COMMENT_SIMILARITY	Belongs to the retinoblastoma protein (RB) family. {ECO:0000305}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Cytoplasm {ECO:0000256|ARBA:ARBA00004496}. Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000256|ARBA:ARBA00004123}.
General Annotations	UP_COMMENT_SUBCELLULAR_LOCATION	Nucleus {ECO:0000269|PubMed:20940255}. Cytoplasm {ECO:0000250|UniProtKB:P13405}. Note=During keratinocyte differentiation, acetylation by KAT2B/PCAF is required for nuclear localization (PubMed:20940255). Localizes to the cytoplasm when hyperphosphorylated (By similarity). {ECO:0000250|UniProtKB:P13405, ECO:0000269|PubMed:20940255}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with HPV E7 protein. {ECO:0000269|PubMed:1316611}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with SV40 large T antigen. {ECO:0000269|PubMed:1316611, ECO:0000269|PubMed:2839300}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with adenovirus E1A protein. {ECO:0000269|PubMed:17974914}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with human cytomegalovirus/HHV-5 proteins UL82 and UL123. {ECO:0000269|PubMed:12626766, ECO:0000269|PubMed:8892909}.
General Annotations	UP_COMMENT_SUBUNIT	(Microbial infection) Interacts with molluscum contagiosum virus protein MC007. {ECO:0000269|PubMed:18701596}.
General Annotations	UP_COMMENT_SUBUNIT	The hypophosphorylated form interacts with and sequesters the E2F1 transcription factor, thereby inhibiting E2F1 transcription (PubMed:20940255, PubMed:8336704). Interacts with heterodimeric E2F/DP transcription factor complexes containing TFDP1 and either E2F1, E2F3, E2F4 or E2F5, or TFDP2 and E2F4. Interacts (when hyperphosphorylated and hypophosphorylated) with PKP3; the interaction inhibits RB1 interaction with and repression of the transcription factor E2F1, potentially via sequestering RB1 to the cytoplasm (By similarity). The unphosphorylated form interacts with EID1, ARID3B, KDM5A, SUV39H1, MJD2A/JHDM3A and THOC1. Interacts with the N-terminal domain of TAF1. Interacts with SNW1, ATAD5, AATF, DNMT1, LIN9, LMNA, KMT5B, KMT5C, PELP1,

Main Category	Category	Terms
		UHRF2 and TMPO-alpha. Interacts with GRIP1 and UBR4. Interacts with ARID4A and KDM5B. Interacts with E4F1 and LIMD1. Interacts with SMARCA4/BRG1 and HDAC1 (By similarity). Interacts with PSMA3 and USP4. Interacts (when methylated at Lys-860) with L3MBTL1. Interacts with CHEK2; phosphorylates RB1. Interacts with CDK1 and CDK2 (By similarity). Interacts with PRMT2. Interacts with CEBPA (PubMed:15107404). P-TEFB complex interacts with RB1; promotes phosphorylation of RB1 (PubMed:12037672). Interacts with RBBP9; the interaction disrupts RB1 binding to E2F1 (By similarity). Interacts with KAT2B/PCAF and EP300/P300 (By similarity). Interacts with PAX5 (PubMed:10197586). Interacts (phosphorylated and unphosphorylated) with BLCAP (PubMed:26986503). May interact with NDC80. {ECO:0000250|UniProtKB:P13405, ECO:0000250|UniProtKB:P33568, ECO:0000269|PubMed:10197586, ECO:0000269|PubMed:10409732, ECO:0000269|PubMed:10446990, ECO:0000269|PubMed:10779342, ECO:0000269|PubMed:10869426, ECO:0000269|PubMed:10888886, ECO:0000269|PubMed:11223246, ECO:0000269|PubMed:11484059, ECO:0000269|PubMed:11571652, ECO:0000269|PubMed:12037672, ECO:0000269|PubMed:12450794, ECO:0000269|PubMed:12466551, ECO:0000269|PubMed:12475961, ECO:0000269|PubMed:12682072, ECO:0000269|PubMed:12695505, ECO:0000269|PubMed:15107404, ECO:0000269|PubMed:15538385, ECO:0000269|PubMed:15542589, ECO:0000269|PubMed:15803180, ECO:0000269|PubMed:15927959, ECO:0000269|PubMed:15949438, ECO:0000269|PubMed:16214886, ECO:0000269|PubMed:16337594, ECO:0000269|PubMed:16360038, ECO:0000269|PubMed:16645588, ECO:0000269|PubMed:17380128, ECO:0000269|PubMed:17996702, ECO:0000269|PubMed:20870719, ECO:0000269|PubMed:20940255, ECO:0000269|PubMed:21952639, ECO:0000269|PubMed:26986503, ECO:0000269|PubMed:7525595, ECO:0000269|PubMed:7935440, ECO:0000269|PubMed:8336704, ECO:0000269|PubMed:8414517, ECO:0000269|PubMed:9858607}.
General Annotations	UP_COMMENT_TISSUE_SPECIFICITY	Expressed in the retina. Expressed in foreskin keratinocytes (at protein level) (PubMed:20940255). {ECO:0000269|PubMed:20940255}.
Identifiers	ENTREZ_GENE_ID	5925
Identifiers	ENSEMBL_GENE_ID	ENSG00000139687
Identifiers	UNIPROT_ID	V9H012_HUMAN
Identifiers	UNIPROT_ID	V9GZJ8_HUMAN
Identifiers	UNIPROT_ID	V9H058_HUMAN
Identifiers	UNIPROT_ID	V9GZQ2_HUMAN
Identifiers	UNIPROT_ID	V9H0A9_HUMAN
Identifiers	UNIPROT_ID	V9H020_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	V9H0D2_HUMAN
Identifiers	UNIPROT_ID	RB_HUMAN
Identifiers	UNIPROT_ID	Q92727_HUMAN
Identifiers	UNIPROT_ID	Q92728_HUMAN
Identifiers	UNIPROT_ID	Q86WG4_HUMAN
Identifiers	UNIPROT_ID	Q7Z5Q6_HUMAN
Identifiers	UNIPROT_ID	Q7KZ59_HUMAN
Identifiers	UNIPROT_ID	Q7KZ60_HUMAN
Identifiers	UNIPROT_ID	Q7KZ61_HUMAN
Identifiers	UNIPROT_ID	Q6LE81_HUMAN
Identifiers	UNIPROT_ID	Q6LE82_HUMAN
Identifiers	UNIPROT_ID	Q6LE79_HUMAN
Identifiers	UNIPROT_ID	Q6DUF0_HUMAN
Identifiers	UNIPROT_ID	Q6LE76_HUMAN
Identifiers	UNIPROT_ID	Q6LE80_HUMAN
Identifiers	UNIPROT_ID	Q6LE77_HUMAN
Identifiers	UNIPROT_ID	Q6LE65_HUMAN
Identifiers	UNIPROT_ID	Q2PMD0_HUMAN
Identifiers	UNIPROT_ID	P78495_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BH52_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BH01_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BH04_HUMAN
Identifiers	UNIPROT_ID	A0AAQ5BH24_HUMAN
Identifiers	UNIPROT_ID	A0A3B3IS71_HUMAN
Identifiers	UNIPROT_ID	A0A385FS48_HUMAN
Identifiers	UNIPROT_ID	A0A385FS38_HUMAN
Identifiers	UNIPROT_ID	A0A385FS41_HUMAN
Identifiers	UNIPROT_ID	A0A385FS40_HUMAN
Identifiers	UNIPROT_ID	A0A385FS25_HUMAN
Identifiers	UNIPROT_ID	A0A385FS65_HUMAN
Identifiers	UNIPROT_ID	A0A385FS12_HUMAN
Identifiers	UNIPROT_ID	A0A385FS44_HUMAN

Main Category	Category	Terms
Identifiers	UNIPROT_ID	A0A385FS33_HUMAN
Identifiers	UNIPROT_ID	A0A2R8Y743_HUMAN
Identifiers	UNIPROT_ID	A0A2R8YFL6_HUMAN
Identifiers	UNIPROT_ID	A0A0X9QC09_HUMAN
Interactions	PUBCHEM	4-fluoro-N-(2,3,4,9-tetrahydro-1H-carbazol-2-yl)benzenesulfonamide
Interactions	PUBCHEM	N-(3-methylcyclopentyl)-2,3,4,9-tetrahydro-1H-carbazol-1-amine
Interactions	PUBCHEM	Ebvaciclib
Interactions	PUBCHEM	US10233188, Example 145
Interactions	PUBCHEM	4-[(4-Furan-3-yl-3-hydroxy-benzylamino)-methylene]-6-iodo-4H-isoquinoline-1,3-dione
Interactions	PUBCHEM	(4Z)-4-[({[1-(3-Furyl)-2-oxo-1,2-dihydropyridin-4-yl]methyl}-amino)methylene]-6-iodoisoquinoline-1,3(2H,4H)-dione
Interactions	PUBCHEM	6-tert-Butyl-4-{[(4-hydroxy-5-propoxy-pyridin-2-ylmethyl)-amino]-methylene}-4H-isoquinoline-1,3-dione
Interactions	PUBCHEM	9-methyl-N-[(4-methylphenyl)methyl]-1,2,3,4-tetrahydrocarbazol-2-amine
Interactions	PUBCHEM	N-benzyl-2,3,4,9-tetrahydro-1H-carbazol-2-amine
Interactions	PUBCHEM	6-methyl-N-[(4-methylphenyl)methyl]-2,3,4,9-tetrahydro-1H-carbazol-2-amine;hydrochloride
Interactions	PUBCHEM	N-benzyl-6-methyl-2,3,4,9-tetrahydro-1H-carbazol-2-amine;hydrochloride
Interactions	PUBCHEM	6-methyl-N-(3-methylcyclopentyl)-2,3,4,9-tetrahydro-1H-carbazol-2-amine;hydrochloride
Interactions	PUBCHEM	6-methyl-N-(3-methylcyclopentyl)-2,3,4,9-tetrahydro-1H-carbazol-2-amine
Interactions	PUBCHEM	N-cyclopentyl-2,3,4,9-tetrahydro-1H-carbazol-2-amine
Interactions	PUBCHEM	4-fluoro-N-(2,3,4,9-tetrahydro-1H-carbazol-1-yl)benzenesulfonamide
Interactions	PUBCHEM	N-[(4-methylphenyl)methyl]-2,3,4,9-tetrahydro-1H-carbazol-2-amine
Interactions	PUBCHEM	N-(3-methylcyclopentyl)-6-nitro-2,3,4,9-tetrahydro-1H-carbazol-1-amine
Interactions	PUBCHEM	Staurosporine
Interactions	PUBCHEM	N-cyclopentyl-2,3,4,9-tetrahydro-1H-carbazol-1-amine
Literature	GENERIF_SUMMARY	K616E in exon 19 (c.1846A>G), an AA insertion in exon 7 (c.684-685insAA), R500G in exon 16 (c.1498A>G), and an A insertion in exon 23 (c.2391-2392insA).

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	novel mutations found in Polish patients with familial and/or bilateral retinoblastoma
Literature	GENERIF_SUMMARY	Suppression of tumorigenicity of rat liver tumor cells by human chromosome 13
Literature	GENERIF_SUMMARY	Tumor suppression by a severely truncated species of retinoblastoma protein
Literature	GENERIF_SUMMARY	Allelic deletions of Rb in urine from bladder cancer patients
Literature	GENERIF_SUMMARY	Activation of caspase-3 and cleavage of Rb are associated with p16-mediated apoptosis in human non-small cell lung cancer cells.
Literature	GENERIF_SUMMARY	Results show that Tax directly interacts with CDK4. The Tax/CDK complex represents an active holoenzyme which capably phosphorylates the Rb protein in vitro and is resistant to repression by the inhibitor p21(CIP).
Literature	GENERIF_SUMMARY	A parent-of-origin effect in two families with retinoblastoma is associated with a distinct splice mutation in the RB1 gene.
Literature	GENERIF_SUMMARY	Three regions of the pRB pocket domain affects its inactivation by human papillomavirus E7 proteins in Hela cells
Literature	GENERIF_SUMMARY	Human telomerase accelerates growth of lens epithelial cells through regulation of the genes mediating RB/E2F pathway
Literature	GENERIF_SUMMARY	Oxidized low density lipoprotein induces the cyclin-dependent kinase inhibitor p21(waf1) and the tumor suppressor Rb.
Literature	GENERIF_SUMMARY	Lack of functional pRb results in attenuated recovery of mRNA synthesis and increased apoptosis following UV radiation in human breast cancer cells.
Literature	GENERIF_SUMMARY	The cyclin D1 high and cyclin E high subgroups of breast cancer
Literature	GENERIF_SUMMARY	The alpha-MSH-induced differentiation of COLO 853 human melanoma cells is associated with decreased pRB phosphorylation and accumulation of cells in the G(1) phase.
Literature	GENERIF_SUMMARY	pRB expression and function are normal in 63 of 66 NHL cases, including 12 of 13 lymphomas with loss of one RB1 allele.There was no association between pRB expression/RB1 copy number and apoptotic fraction.
Literature	GENERIF_SUMMARY	Expression of RB C pocket fragments in HSF induces delayed cell cycle progression and sensitizes to apoptosis upon cellular stresses.
Literature	GENERIF_SUMMARY	The spectrum and frequencies of RB1 structural defects were studied in tumors and peripheral blood lymphocytes of patients with various forms of retinoblastoma

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Localization and phosphorylation kinetics of this protein correlate with the cellular phenotype of cultured breast adenocarcinoma cells.
Literature	GENERIF_SUMMARY	Low-penetrance retinoblastoma due to exons 24 and 25 deletions in the Rb1 gene
Literature	GENERIF_SUMMARY	Review. Rb suppresses tumor formation through its multiple biological activities. A theme throughout its multiple cellular functions is its central role in controlling activities that involve chromatin remodeling. Rb may control global genome fluidity.
Literature	GENERIF_SUMMARY	Gene aberrations at chromosome 13 are involved in the progression of laryngeal squamous cell carcinoma; results provided further evidence for the putative role of the RB1 gene alterations in the metastatic process
Literature	GENERIF_SUMMARY	Data show that the retinoblastoma protein specifically activates transcription of the survival gene bcl-2 in epithelial cells but not in NIH 3T3 mesenchymal cells.
Literature	GENERIF_SUMMARY	Aberrant methylation inactivating RB1 was detected in 14 (27%) tumors.Complex testing for RB1 mutations, loss of heterozygosity, and functional inactivation of the two genes revealed a molecular defect in at least one allele in 51 (98%) tumors.
Literature	GENERIF_SUMMARY	Activation of cyclin D1-Cdk4 and Cdk4-directed phosphorylation in diabetic mesangial hypertrophy
Literature	GENERIF_SUMMARY	novel mutations in Mexican patients with retinoblastoma
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	pRB has a role in eye cancer [review]
Literature	GENERIF_SUMMARY	data suggest that hypophosphorylated Rb is anchored in the nucleus by the interaction of pocket C with LAP2alpha-lamin A/C complexes
Literature	GENERIF_SUMMARY	The relationships and interactions between p53, Rb and bcl-2 immunostaining, clinical parameters and response to cisplatin-based chemotherapy were evaluated in the present study.
Literature	GENERIF_SUMMARY	inactivation of both p16(INK4a) and pRb is associated with immortalization of human cells including fibroblasts and epithelial cells and telomerase-positive cells and ALT-positive cells
Literature	GENERIF_SUMMARY	Results describe the relationship between Helicobacter pylori (H.pylori) infection and the expressions of the p53, Rb, c-myc, bcl-2 and hTERT mRNA in a series of diseases from chronic gastritis to gastric cancer.
Literature	GENERIF_SUMMARY	Rb and its inactive phospho-isoform exhibit distinct expression patterns in Parkinson disease neurons most commonly associated with the disease, as well as in neurons elsewhere in the brain that also contribute to disease progression or symptoms.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Expression of p57kip2, Rb protein and PCNA and their relationships with clinicopathology in human pancreatic cancer.
Literature	GENERIF_SUMMARY	RB1 CpG island hypermethylation is a common epigenetic event that is associated with the development of malignant nervous system tumours.
Literature	GENERIF_SUMMARY	first preimplantation genetic diagnosis of hereditary retinoblastoma using microsatellite markers of this protein
Literature	GENERIF_SUMMARY	p55gamma binds to Rb and modification of this association can lead to cell cycle arrest
Literature	GENERIF_SUMMARY	retinoblastoma protein has cyclin D1-inducing activity that is abolished by adenovirus E1A and that involves multiple pocket sequences that are independently involved in cyclin D1 activation
Literature	GENERIF_SUMMARY	report shows hypophosphorylation of the retinoblastoma family proteins induced by H2O2 was because of the activity of protein phosphatase 2A and pRb dephosphorylation may induce an intra-S-phase response that leads to a reduced rate of DNA synthesis
Literature	GENERIF_SUMMARY	We found that E2F1 was present at most of the CpG islands bound by pRb, independent of the phase of the cell cycle, our data suggest that the majority of DNA-bound pRb is recruited to E2F target promoters during both G(0)/G(1) and S phases.
Literature	GENERIF_SUMMARY	Senescence-associated heterochromatic foci formation coincides with the recruitment of heterochromatin proteins and the retinoblastoma (Rb) tumor suppressor to E2F-responsive promoters and is associated with the stable repression of E2F target genes.
Literature	GENERIF_SUMMARY	EBV LMP1 blocks p16INK4-RB pathway by promoting nuclear export of E2F-4 and E2F-5.
Literature	GENERIF_SUMMARY	These results indicate that PAI-2 may enhance Rb's tumor suppressor activity and suggest a potential therapeutic role for PAI-2 against HPV-transformed lesions.
Literature	GENERIF_SUMMARY	C/EBPepsilon interacts with Rb and E2F1 during granulocytic differentiation
Literature	GENERIF_SUMMARY	p21WAF1/CIP1, pRB, Bax and NF-kappaB have roles in induction of growth arrest and apoptosis by resveratrol in tumor cell lines
Literature	GENERIF_SUMMARY	The survival data on 129 glioblasstoma patients were correlated with the results of a detailed analysis of this gene.
Literature	GENERIF_SUMMARY	RB1CC1 induces the expression of RB1, especially of underphosphorylated forms, then suppresses cell cycle progression in human neoplastic cells
Literature	GENERIF_SUMMARY	cell lines lacking an active RB gene are more resistant

Main Category	Category	Terms
		than cell lines with an active RB gene
Literature	GENERIF_SUMMARY	kinetic framework of RB tumor suppressor action in transcriptional repression and cell cycle regulation
Literature	GENERIF_SUMMARY	Rb and ASC-2 have roles in androgen receptor transactivation
Literature	GENERIF_SUMMARY	Androgens repress Bcl-2 expression via activation of RB in prostate cancer cells.
Literature	GENERIF_SUMMARY	results establish that tumor necrosis factor alpha targets insulin-like growth factor-I induced E2F-transcription facor 1 synthesis, leading to inhibition of accumulation in cyclin A and hyperphosphorylation of RB protein
Literature	GENERIF_SUMMARY	Rb is present in rat and human islets, and overexpression of cyclin D(1)/cdk-4 led to strikingly enhanced Rb phosphorylation.
Literature	GENERIF_SUMMARY	Fortythree novel mutations found in RB1, the mutational spectrum displayed a surprising amount of splice mutations and large deletions.
Literature	GENERIF_SUMMARY	Calcitrio may signal cell differentiation of HL 60 cells which in turn regulates expresson of RB protein.
Literature	GENERIF_SUMMARY	Rb negatively regulates p38 activation, leading to decreased MMP-1 secretion in rheumatoid arthritis synovial fibroblasts.
Literature	GENERIF_SUMMARY	pRb and p107 do not constitute the last control point for extracellular factors during G(1)-phase progression, and they functionally separate the requirements for serum and cell anchorage in terms of involved cell cycle components.
Literature	GENERIF_SUMMARY	role of Rb family memebers in stability of repressor complexes at promoters
Literature	GENERIF_SUMMARY	The overall expression of PCNA in all stages of development was higher than pRb1 expression.
Literature	GENERIF_SUMMARY	The mechanisms for pRb tumor suppression in the epithelia of two distinct tissues, mammary gland and brain, are indistinguishable. (review)
Literature	GENERIF_SUMMARY	Results suggest that acetylation regulates the differentiation-specific function(s) of retinoblastoma tumor-suppressor protein.
Literature	GENERIF_SUMMARY	Rb regulates apoptotic genes differently in human and in mouse cells
Literature	GENERIF_SUMMARY	A non-cdk8-associated cellular pool of cyclin C combines with cdk3 to stimulate pRb phosphorylation at S807/811 during the G0/G1 transition, and this phosphorylation is required for cells to exit G0 efficiently.
Literature	GENERIF_SUMMARY	Data show that Oct-1 occupies the endogenous HLA-

Main Category	Category	Terms
		DRA promoter when the HLA-DRA promoter is inactive in retinoblastoma protein-defective cells.
Literature	GENERIF_SUMMARY	Rb is absent or phosphorylated in most anaplastic large cell lymphoma cell lines and tumors and absence of Rb expression is associated with better clinical outcome in patients with ALCL.
Literature	GENERIF_SUMMARY	A novel Cdk4 docking motif has been defined within a stretch of 19 amino acids from the C-terminal domain of the Rb protein that are essential for Cdk4 binding.
Literature	GENERIF_SUMMARY	In mice, Rb/p107 loss has cell-specific effects in retina
Literature	GENERIF_SUMMARY	RB can repress U6 snRNA gene transcription at critical steps subsequent to RNA polymerase III recruitment
Literature	GENERIF_SUMMARY	Rb acts as a link between apoptotic and proliferative pathways by interacting with distinct kinases and occupying different promoters
Literature	GENERIF_SUMMARY	Hereditary retinoblastoma is an autosomal dominant disorder caused by mutations in the RB1 gene [Review]
Literature	GENERIF_SUMMARY	Generation of Rb pathway lesions in normal and transformed cells produces aberrant Mad2 expression and mitotic defects leading to aneuploidy, such that elevated Mad2 contributes directly to these defects
Literature	GENERIF_SUMMARY	Immortalization of MRC5hTERT cells was associated with repression of the cyclin-dependent kinase inhibitor p16INK4a and up-regulation of pRB.
Literature	GENERIF_SUMMARY	herpesvirus saimiri (HVS) ORF73 binds both p53 and pRb in vitro and in vivo, colocalizes with p53 in T cells infected with HVS, and in cells overexpressing both ORF73 and p53, and adversely influences pRB/E2F and p53 transcriptional regulation
Literature	GENERIF_SUMMARY	RB/E2F signaling pathway activation is required for the modulation of hepatitis C virus core protein-induced cell growth in liver and non-liver cells
Literature	GENERIF_SUMMARY	Results demonstrate that p16ink4A-mimicking peptides can induce apoptosis in vitro and inhibit tumor growth in vivo in p16-defective, retinoblastoma protein-expressing human melanoma cells.
Literature	GENERIF_SUMMARY	p21(WAF1), acting through the phosphorylation of pRb, regulates whether 2BS cells cease to proliferate and become senescent but resistant to apoptosis, or whether they accelerate proliferation while becoming more susceptible to apoptotic stimuli
Literature	GENERIF_SUMMARY	dynamic equilibrium between CDKs and PP2A modulates phosphorylation of pRB
Literature	GENERIF_SUMMARY	Constitutional RB1 germ-line mutations were studied in a series of Argentine families.
Literature	GENERIF_SUMMARY	the central acidic domain of MDM2 is critical in inhibition of retinoblastoma-mediated suppression of

Main Category	Category	Terms
		E2F and cell growth
Literature	GENERIF_SUMMARY	Raf-1 links mitogenic signaling to Retinoblastoma Protein and that disruption of this interaction could aid in controlling proliferative disorders
Literature	GENERIF_SUMMARY	Aberrant expression of pRb and p16, alone and in combination, heralds poor prognosis in patients with CRC.
Literature	GENERIF_SUMMARY	Evaluation of phosphorylated retinoblastoma protein in melanocytic tumors could become a helpful adjunct in clinicopathological routine.
Literature	GENERIF_SUMMARY	p16-Rb pathway plays important role in tumor progression and prognosis in vertical growth phase melanomas
Literature	GENERIF_SUMMARY	phosphorylation promoted by HPV16 E6 protein
Literature	GENERIF_SUMMARY	These results suggest that Mdm2 regulates function of retinoblastoma gene product (pRB) via ubiquitin-dependent degradation of pRB.
Literature	GENERIF_SUMMARY	a model whereby Tax targets Rb to the proteasome by acting as a molecular bridge bringing Rb into contact with the proteasome for degradation
Literature	GENERIF_SUMMARY	the derepression of Rb-E2F-regulated genes leads to apoptosis through inactivation of focal adhesion kinase and activation of caspase-8
Literature	GENERIF_SUMMARY	Rb, p21Cip1, and PCNA have roles in cAMP-mediated inhibition of DNA replication and S phase progression
Literature	GENERIF_SUMMARY	BMP-2 inhibits DHT-induced growth of LNCaP cells through a decrease in E2F protein expression and suppression of E2F activity by hypophosphorylation of Rb
Literature	GENERIF_SUMMARY	Mutation of p16 and Rb genes might be correlated with progression of gallbladder carcinoma
Literature	GENERIF_SUMMARY	Rb and p53 have roles in progression of primary non-small cell lung carcinoma
Literature	GENERIF_SUMMARY	a functional interaction between pRB and HIF-1alpha is confirmed by showing that HIF-1alpha reverses the transcription repressor function of pRB.
Literature	GENERIF_SUMMARY	an alternatively translated form of pRb is expressed during normal myelopoiesis and in tumour-associated macrophages
Literature	GENERIF_SUMMARY	experiments suggest that both catalytic and non-catalytic functions of cyclin D1 can collaborate to inactivate pRb-mediated cell cycle exit
Literature	GENERIF_SUMMARY	Rb interacts with the NeuroD1/Beta2 transcription factor and activates the POMC gene
Literature	GENERIF_SUMMARY	Rb is a coactivator of NeuroD1/Beta2 on the POMC

Main Category	Category	Terms
		promoter
Literature	GENERIF_SUMMARY	Suppression of pRb may be characteristic for papillary thyroid carcinomas with a tendency to early metastasizing.
Literature	GENERIF_SUMMARY	Mitf-mediated activation of p21Cip1 expression and consequent hypophosphorylation of Rb1 will contribute to cell cycle exit and activation of the differentiation programme
Literature	GENERIF_SUMMARY	Data showed that disabling of the Rb pathway was frequent event in oral carcinoma.
Literature	GENERIF_SUMMARY	an early event in neuronal cell death is p25/Cdk5-mediated retinoblastoma phosphorylation
Literature	GENERIF_SUMMARY	Dehydroepiandrosterone diminished the levels of phosphorylated retinoblastoma protein.
Literature	GENERIF_SUMMARY	Knockdown of the gene in colorectal carcinoma cells caused resistance to antitumor agents.
Literature	GENERIF_SUMMARY	Rb acts as a coactivator of HNF-4 and is recruited to the alpha1-AntiTrypsin promoter during the enterocyte differentiation of Caco2 cells
Literature	GENERIF_SUMMARY	Rb enhances the POMC promoter activation by the CRH hormone in mouse AtT-20 corticotroph cells
Literature	GENERIF_SUMMARY	Rb interacts with and acts as a potentiator of the p160 transcriptional coactivator SRC-2/GRIP1/TIF-2
Literature	GENERIF_SUMMARY	Rb interacts with the nuclear receptors
Literature	GENERIF_SUMMARY	Rb potentiates the activity of the SRC-2/GRIP1/TIF-2, SRC-1 and SRC-3 coactivators on the nuclear receptors
Literature	GENERIF_SUMMARY	the RB1 gene is conserved in human and primates
Literature	GENERIF_SUMMARY	RB gene might be down-regulated at the transcriptional level in human testicular seminoma cells.
Literature	GENERIF_SUMMARY	Smad7 acts to functionally inactivate RB and de-repress E2F without blocking the activation of TbetaRI and the nuclear translocation of Smad2/3, allowing TGF-beta1 to exert effects in a cancer cell that is resistant to TGF-beta1-mediated growth inhibition
Literature	GENERIF_SUMMARY	p130/p107/p105Rb has a role in transcriptional repression in DNA-damage-induced cell-cycle exit at G2
Literature	GENERIF_SUMMARY	nuclear import conferred by the RB nuclear localization sequence has distinct properties, in part due to the affinity of its interaction with IMPalpha
Literature	GENERIF_SUMMARY	The Aberrant expression of Rb Genes were determined in bone marrow samples of children with de novo B-lineage (n=170) and T-lineage (n=25) acute lymphoblastic leukemia (ALL).
Literature	GENERIF_SUMMARY	RB1 gene has a major role in the development of

Main Category	Category	Terms
		human vestibular schwannomas, increased levels of RB1 mRNA, total pRb and the phosphorylated form of it suggests that RB1 gene in these tumors may have anti-apoptotic function
Literature	GENERIF_SUMMARY	Germline RB1 mutations in 77 out of 85 bilateral RB patients (91%), 7 out of 10 familial unilateral (70%), and 6 out of 85 unilateral patients with no family history of RB (7%), were identified.
Literature	GENERIF_SUMMARY	developmental expression of RB, p130 and p107 in mouse and human retina
Literature	GENERIF_SUMMARY	Deregulation of both pRb and p53 pathways is associated with malignant transformation in oral tumorigenesis
Literature	GENERIF_SUMMARY	Modulates the subcellular localization of BAG-1 in colorectal tumor cell susrvival.
Literature	GENERIF_SUMMARY	Gross deletions and insertions in the RB1 gene are associated with retinoblastoma.
Literature	GENERIF_SUMMARY	data suggest that loss of Rb creates strong selective pressure, via DSB accumulation, for inactivating p53 mutations and that E2F1 contributes to the genetic instability associated with transformation and tumorigenesis
Literature	GENERIF_SUMMARY	data show that E2F1 has potential binding activity to the retinoblastoma control element and a different transcriptional regulation pathway which cooperates with the retinoblastoma tumor suppressor protein
Literature	GENERIF_SUMMARY	results show that phosphorylation site-mutated RB exerts disparate effects on apoptotic response to different stimuli, and that cell cycle arrest does not always associate with resistance to apoptosis
Literature	GENERIF_SUMMARY	pRB degradation involved in gastric tumor cell apoptosis is p53-independent
Literature	GENERIF_SUMMARY	Methylation in the p16 promoter region is biologically significant, being associated with phosphorylation of pRb and cell growth in human hepatocellular carcinoma cells
Literature	GENERIF_SUMMARY	RB expression may be a useful biomarker for assessing the risk of developing esophageal cancer.
Literature	GENERIF_SUMMARY	DGKzeta may act in vivo as a downstream effector of pRB to regulate nuclear levels of diacylglycerol and phosphatidic acid
Literature	GENERIF_SUMMARY	Sars virus 7a protein expression was correlated with phosphorylation of retinoblastoma (Rb) protein at ser795 and ser809/811 and prevention of cell cycle progression at the G0/G1 phase.
Literature	GENERIF_SUMMARY	Review. RB1 is a marker of retinoblastoma. It maps to 13q14.2 & encodes a 110-kDa nuclear phosphoprotein, cell-cycle-regulated phosphorylation of which plays a

Main Category	Category	Terms
		major role in cell proliferation control. Most point mutations are in exons 3, 8, 18 & 20.
Literature	GENERIF_SUMMARY	Down-regulation of the retinoblastoma tumor suppressor by the hepatitis C virus NS5B RNA-dependent RNA polymerase.
Literature	GENERIF_SUMMARY	These findings suggest that overexpression of Mdm2 can perturb a RB pathway regardless of the p53 gene status, promoting carcinogenesis.
Literature	GENERIF_SUMMARY	EBNA3C forms a stable complex with Rb in cells when the proteasome machinery is inhibited and interacts with Rb in vitro, mapping to a conserved domain at the terminus of EBNA3C
Literature	GENERIF_SUMMARY	Clear need in human melanoma cell lines to disrupt both RB and p53 pathways and recurrent mechanisms which play into the unique genetic vulnerabilities of this tumor type.
Literature	GENERIF_SUMMARY	Findings explain the requirement of the Rb C-terminal domain for high-affinity E2F binding and growth suppression, and establish a mechanism for the regulation of Rb-E2F association by phosphorylation.
Literature	GENERIF_SUMMARY	Rb binding to HPV 18 E7 is modified by transglutaminase 2 with different polyamines
Literature	GENERIF_SUMMARY	RB1 was not implicated, suggesting the participation of another tumor suppressor gene in this region.
Literature	GENERIF_SUMMARY	Possible role for the G-quadruplex in control of DNA synthesis at the 5' end of the retinoblastoma susceptibility gene.
Literature	GENERIF_SUMMARY	Our data suggests that RB1 may play a role in colorectal tumorigenesis through functional regulation of the transcript and protein rather than through its tumour suppressor role by gene inactivation.
Literature	GENERIF_SUMMARY	These findings suggest heterogeneous abnormalities of CCND and RB in cutaneous T-cell lymphomas, in which dysregulated CCND and RB1 may lead to impaired cell cycle control.
Literature	GENERIF_SUMMARY	Differences in oncoprotein expression between endometriotic and adenomyotic tissues provide further evidence that the pathogenesis of endometriosis is different from that of adenomyosis.
Literature	GENERIF_SUMMARY	determination that Rb is not activated in hTERT immortalization of urothelial cells
Literature	GENERIF_SUMMARY	RB1 gene was methylated in seven samples (14%) and each patient had refractory anemias(RA).
Literature	GENERIF_SUMMARY	A molecular study of first and second RB1 mutational hits in retinoblastoma patients was performed.
Literature	GENERIF_SUMMARY	intracellular distribution of nucleolin in epithelial cells is Rb-dependent, and an altered nucleolin localization in

Main Category	Category	Terms
		human cancerous tissues results from a loss of Rb
Literature	GENERIF_SUMMARY	when nontransformed CV1 epithelial cells and Hs578T breast cancer cells are treated with cytosine arabinoside,Thr-821 of Rb is rapidly dephosphorylated,with dissociation of the PP1 regulatory subunit PNUTS (phosphatase nuclear targeting subunit)from PP1.
Literature	GENERIF_SUMMARY	facets of RB protein function in cell cycle control in retinoblastoma (review)
Literature	GENERIF_SUMMARY	Membrane depolarization may stimulate cellular proliferation by augmenting the expression of cyclin E leading to increases in Cdk2 activity and RB phosphorylation in a neuroblastoma cell line.
Literature	GENERIF_SUMMARY	These findings reveal that phosphorylation of threonine-373 by CDK2-cyclin E represent a potentially crucial event in the inactivation of the pRb protein.
Literature	GENERIF_SUMMARY	Relationship between Rb and Ini1 in tumor suppression indicate that Ini1 plays a role in maintaining the morphologic and functional differentiation of corticotrophic cells.
Literature	GENERIF_SUMMARY	The transformation of melanocytes to melanoma cells is characterised by abnormal proliferation due to cell cycle alterations. This occurs through alterations in the retinoblastoma (Rb) and p53 tumour suppressor pathways.
Literature	GENERIF_SUMMARY	nonphosphorylated p68 inhibited the stimulation of pol-alpha activity by hyperphosphorylated retinoblastoma protein, suggesting that p68 might impede the association of ppRb with p180
Literature	GENERIF_SUMMARY	The antiproliferative effects of Lf can likely be attributed to the elevated levels of hypophosphorylated Rb.
Literature	GENERIF_SUMMARY	Determination of intraepithelial height of immunohistochemical staining for p16, p53, pRb, and Ki-67 in upper aerodigestive tract lesions with reference to degree of dysplasia
Literature	GENERIF_SUMMARY	these studies highlight p38 MAPK, HBP1, and RB as important components for a premature-senescence pathway with possible clinical relevance to breast cancer.
Literature	GENERIF_SUMMARY	CDCA4 participates in the regulation of cell proliferation, mainly through the E2F/retinoblastoma protein pathway
Literature	GENERIF_SUMMARY	p53 and pRB can be sumoylated by SUMO-2/3 in vivo, and such modification of p53 and pRB may play roles in premature senescence and stress response
Literature	GENERIF_SUMMARY	Results uncover an unexpected role for the p16(INK4a)-Rb pathway and provide a new insight into how senescent cell-cycle arrest is enforced.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Data suggest that in Ehrlich ascites tumor cells, growth inhibition by 1'-acetoxychavicol acetate involves decreased Rb and p27(kip1) phosphorylation and increased nuclear localization of p27(kip1), dependent on cellular thiol status.
Literature	GENERIF_SUMMARY	significant cytoplasmic mislocalization of ordinarily nuclear RB in cells harboring Cdk4 mutations
Literature	GENERIF_SUMMARY	Promoter methylation of retinoblastoma Protein is associated with malignant fibrous histiocytomas
Literature	GENERIF_SUMMARY	mutation of Rb gene is frequent in gastric carcinoma; expression of altered protein inversely correlates with tumor invasion
Literature	GENERIF_SUMMARY	The pRb level was higher than in normal larynxes, whereas laryngeal cancer presented the lowest levels.
Literature	GENERIF_SUMMARY	RB pathway is a critical determinant of tumorigenic proliferation and differential therapeutic response and may represent a critical basis for directing therapy in the treatment of breast cancer.
Literature	GENERIF_SUMMARY	Rb induces a proliferative arrest and regulates Brn-2 expression in retinoblastoma cells
Literature	GENERIF_SUMMARY	Human papillomavirus E7 repression in cervical carcinoma cells initiates a transcriptional cascade driven by the retinoblastoma family, resulting in senescence.
Literature	GENERIF_SUMMARY	The growth suppression function of RB requires a conserved A/B domain that contains two distinct binding pockets. At the A/B interface is a binding site for the C-terminal trans-activation domain of E2F. The B-domain has a binding site for the LxCxE motif
Literature	GENERIF_SUMMARY	aberrant hypermethylation of the key cell cycle regulatory genes occurs at a relatively high frequency in pituitary adenomas, especially in RB1 pathway genes with promoter hypermethylation of the p16(INK4a) gene being the most common deregulation
Literature	GENERIF_SUMMARY	Sequence deletion or mutation or gene deletion in retinoblastoma patients in Argentina.
Literature	GENERIF_SUMMARY	This study demonstrates that ARF plays a direct role in regulation of Rb and suggests that inactivation of ARF may lead to defects in both p53 and Rb pathways in human cancer development.
Literature	GENERIF_SUMMARY	This is the first report of a deep intronic mutation in RB1 and is a proof of concept that some undetected RB1 mutations should be investigated at the cDNA level.
Literature	GENERIF_SUMMARY	Presence of LOH at the RB1 gene locus and the increased levels of RB1 RNA and protein and increased percentage of hyperphosphorylated form of pRb are indicative of an overall deregulation of pRb pathway in human brain tumors.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	RB1 may be inactivated in aggressive mantle-cell lymphoma by intragenic deletions.
Literature	GENERIF_SUMMARY	TGF-beta1 increased retinoblastoma protein phosphorylation at both Ser807/811 and Ser780 in airway smooth muscle cells
Literature	GENERIF_SUMMARY	Radiation-induced somatic mutations are not observeded in radiation-induced sarcoma, although the gene was activated in sarcoma.
Literature	GENERIF_SUMMARY	Phosphorylation of pRB at Ser612 by Chk1/2 leads to a complex between pRB and E2F-1 after DNA damage.
Literature	GENERIF_SUMMARY	LOH of the retinoblastoma gene in is not shown to have a role in progression of osteosarcoma
Literature	GENERIF_SUMMARY	SARS-CoV 3a protein, through limiting the expression of cyclin D3, may inhibit Rb phosphorylation, which in turn leads to a block in the G1 phase of the cell cycle and an inhibition of cell proliferation
Literature	GENERIF_SUMMARY	Telomere dysfunction and inactivation of the rb pathway might play a role for pyrothorax-associated lymphoma
Literature	GENERIF_SUMMARY	these analyses demonstrate that cyclin A1 exerts antiapoptotic functions by interacting with retinoblastoma and Ku proteins in leukemia cells.
Literature	GENERIF_SUMMARY	Abnormal expression of the pRB protein may be implicated in the process of esophageal carcinogenesis.
Literature	GENERIF_SUMMARY	In Ishikawa H cells that model type I endometrial cancer in the loss of PTEN and RB1, re-expressing PTEN and RB1 increased the apoptotic and G1 phases and decreased the S and G2-M phases, which further sensitize the cells to gefitinib.
Literature	GENERIF_SUMMARY	sufficient Rb levels are important for the cytotoxic and anticlonogenic effects of ErPC3 at levels below the IC(50), but higher concentrations of ErPC3 are less dependent on Rb status
Literature	GENERIF_SUMMARY	A microtuble-facilitated nuclear import pathway for RB1 is described.
Literature	GENERIF_SUMMARY	Promoter hypermethylation of RB1 was observed.
Literature	GENERIF_SUMMARY	Inactivation of retinoblastoma protein is associated with head and neck squamous cell carcinomas
Literature	GENERIF_SUMMARY	Down-regulation of Rb Protein is associated with defects in erythroblast islands
Literature	GENERIF_SUMMARY	Results identify BNIP3 as a key regulator of hypoxia-induced autophagy and suggest a novel role for the RB tumor suppressor in preventing nonapoptotic cell death by limiting the extent of BNIP3 induction in cells.
Literature	GENERIF_SUMMARY	Epstein-Barr virus-encoded latent membrane protein

Main Category	Category	Terms
		1inhibited p16(INK4A) expression, promoted phosphorylation of p105 Rb and upregulated E2F1 expression and overexpression of E2F1 alone was sufficient to upregulate telomerase activity.
Literature	GENERIF_SUMMARY	the HPV16 E7-associated cullin 2 ubiquitin ligase complex contributes to aberrant degradation of the pRB tumor suppressor in HPV16 E7-expressing cells.
Literature	GENERIF_SUMMARY	RbAp48-mediated transformation of HPV16 is probably because of the regulation by RbAp48 of tumor suppressors retinoblastoma and p53, apoptosis-related enzymes caspase-3 and caspase-8, E6, E7, cyclin D1 (CCND1), and c-MYC.
Literature	GENERIF_SUMMARY	RB depletion dramatically alters the cellular response to therapeutic intervention in prostate cancer cells
Literature	GENERIF_SUMMARY	RB reconstitution into RB-deficient NSCLC lines establishes regulation of certain RB/E2F target genes and restores G1 arrest mechanisms
Literature	GENERIF_SUMMARY	correlation between RB1 gene deletion and mammary gland carcinoma
Literature	GENERIF_SUMMARY	Aberrant methylation of multiple genes (E-cadherin, estrogen receptor, RB1 , p16, p15, p14, and MGMT) is involved in gastric carcinogenesis.
Literature	GENERIF_SUMMARY	retinoblastoma protein downregulation is involved in the enhanced cytotoxicity of 4-hydroxytamoxifen plus mifepristone combination therapy versus antiestrogen monotherapy of human breast cancer
Literature	GENERIF_SUMMARY	ectopically BRCA2-expressing cells have different intracellular levels of Aurora A, Aurora B, p21, E2F-1, and pRb, suggesting a BRCA2-mediated suppression of polyploidy via stabilization of the checkpoint proteins levels
Literature	GENERIF_SUMMARY	Aberrations of the p53, Rb and p27 pathways are associated with aggressive clinical behavior in DLBCL.
Literature	GENERIF_SUMMARY	homozygous inactivation of P2RY5 was antecedent to the loss of RB1 during tumor development
Literature	GENERIF_SUMMARY	KAP1 contributes to the negative regulation of E2F1 and may serve as a partial backup to prevent E2F1-mediated apoptosis in the absence of pRb
Literature	GENERIF_SUMMARY	It is recorded that borealin is a cell cycle regulator, down-regulated in response to p53/Rb-signaling, and up-regulated in many types of cancerous tissues.
Literature	GENERIF_SUMMARY	Retinoblastoma deficiency increases chemosensitivity in lung cancer.
Literature	GENERIF_SUMMARY	High mobility group protein A1 (HMGA1)disturbs retobblastoma protein-mediated cell arrest, suggesting a negative control of RB by HMGA1.
Literature	GENERIF_SUMMARY	Somatic down-regulation of mismatch repair proteins in

Main Category	Category	Terms
		nodular-trabecular muscle-invasive bladder urothelial carcinomas results in RB1/NF1 microsatellite abnormalities, correlating with higher cellular turnover and longer survival.
Literature	GENERIF_SUMMARY	There is more frequent p16 hypermethylation in mantle cell lymphoma and p15 or Rb1 hypermethylation in follicular lymphoma
Literature	GENERIF_SUMMARY	Myc down-regulation is a mechanism to activate the Rb pathway in STAT5A-induced senescence
Literature	GENERIF_SUMMARY	Human herpesvirus 6A (HHV-6A) and HHV-6B infection was associated with significant reduction of E2F1/Rb complexing, and Rb protein was dephosphorylated early postinfection.
Literature	GENERIF_SUMMARY	compromise of either p53 or Rb pathways during melanocyte transformation leads to up-regulation of survivin expression in melanoma
Literature	GENERIF_SUMMARY	Ten novel mutations were identified, including four single base substitutions, four small deletions and two small duplications. Predominantly gene-inactivating mutations, i.e. single-base non-sense mutations and splice site mutations.
Literature	GENERIF_SUMMARY	mechanisms other than RB1 gene changes may lead to retinoblastoma because not all cases of retinoblastoma show gene alterations
Literature	GENERIF_SUMMARY	A novel molecular mechanism for pRb inactivation by a viral oncoprotein.
Literature	GENERIF_SUMMARY	Calpain emerges as a central player in E7-mediated degradation of Rb
Literature	GENERIF_SUMMARY	underline the importance of Protein phosphatase 2A subunit PR70-Ca(2+) interaction in the signal transduction mechanisms triggered by redox imbalance and leading to pRb dephosphorylation
Literature	GENERIF_SUMMARY	The crystal structure of the Rb N-terminal domain (RbN) was determined.
Literature	GENERIF_SUMMARY	Study described the use of minigene constructs to study the oncogenic character of intronic RB1 variants detected during mutational screening and show the utility of this approach to ascertain the oncogenic nature of unique RB1 intronic variants.
Literature	GENERIF_SUMMARY	These data shed more light on the molecular biology of meningioma cells and suggest that survivin and proteins of the RB pathway could play a determinant role in the development and the treatment of meningiomas.
Literature	GENERIF_SUMMARY	RB inactivation, via aberrant nucleocytoplasmic transport, may disrupt normal cell differentiation programs and accelerate the cancer process.
Literature	GENERIF_SUMMARY	defines a novel family of cellular factors linked to cell

Main Category	Category	Terms
		proliferation and pRb/E2F cell cycle pathways in humans, fish, and nematodes
Literature	GENERIF_SUMMARY	analyzed RNA from retinoblastoma patients and unaffected carriers with various RB1 gene mutations to determine the patterns of missplicing and associations with phenotypic expression
Literature	GENERIF_SUMMARY	In this review, retinoblastoma protein acts along with Skp2 to regulate signal transduction in the cell cycle.
Literature	GENERIF_SUMMARY	Loss of both RB1 tumor suppressor gene alleles initiates quiescent RB1(-/-) retinomas with low level genomic instability and high expression of the senescence-associated proteins p16(INK4a) and p130.
Literature	GENERIF_SUMMARY	These findings indicate that Epstein-Barr virus C promoter (Cp) is a cell cycle-regulated promoter that is under the control of Rb and the histone demethylase LSD1 in multiple latency types.
Literature	GENERIF_SUMMARY	Findings strongly demonstrate that retinoblastoma (RB) and cyclin-dependent kinase 2 (CDK2) on one side and cytokeratin 8 (CK8) and epidermal growth factor receptor 2 (HER2) on the other may affect the clinical course of the disease in 56% of patients.
Literature	GENERIF_SUMMARY	The deletion of the RB1 gene plays an important role in the development of acute myelocytic leukemia.
Literature	GENERIF_SUMMARY	Our data suggest that p107 and p130, but not pRb, and the repressor E2F, but not activator E2Fs, play a critical role in regulating E2F-responsive gene expression.
Literature	GENERIF_SUMMARY	The morphology and cell cycle proteins immunoexpression of the novel probable preinvasive lesion - bronchiolar columnar cell dysplasia (BCCD), is decribed.
Literature	GENERIF_SUMMARY	Cdk6 up-regulation in TDP-43-depleted cells is accompanied by an increase in phosphorylation of two of its major targets, the retinoblastoma protein pRb and pRb-related protein pRb2/p130.
Literature	GENERIF_SUMMARY	Analysis of the pattern of expression of these biomolecules showed increased p16-positive phenotypes and decreased cyclin D1- and pRB-positive phenotype among the invasive tumors compared to low-grade CIN lesions
Literature	GENERIF_SUMMARY	We concluded from these studies that both UL97 kinase activity and the LxCxE retinoblastoma protein (RB) binding motif are required for the phosphorylation and stabilization of RB in infected cells.
Literature	GENERIF_SUMMARY	Rb antagonizes gankyrin to inhibit MDM2-mediate p53 ubiquitination in cancer cells and suggest that the status of both p53 and Rb is important for efficacy of cancer chemotherapy.
Literature	GENERIF_SUMMARY	Results suggest that Bin1 gene suppression caused by oncogenic E1A via Rb/E2F1 inactivation is an essential

Main Category	Category	Terms
		step in cell cycle progression promoted by c-Myc, and subsequently, E1A transformation.
Literature	GENERIF_SUMMARY	reduced expression of PNUTS leads to activation of Rb-phosphatase and caspase-mediated apoptosis
Literature	GENERIF_SUMMARY	The absence of pRB expression renders human breast cancer cells more sensitive to 5-FU and methotrexate and predicts a good clinical outcome for patients treated with adjuvant chemotherapy
Literature	GENERIF_SUMMARY	Increased retinoblastoma 1 expression was associated with invasive adenocarcinoma of the prostate
Literature	GENERIF_SUMMARY	tumorigenesis of RMS may be associated with retinoma gene alteration
Literature	GENERIF_SUMMARY	MRPS18-2 binds to both hypo- and hyperphosphorylated forms of Rb protein specifically. This binding targets the small pocket of pRb, which is a site of interaction with E2F1.
Literature	GENERIF_SUMMARY	Human breast cancer-associated fibroblasts show CAV1 down-regulation and RB1 tumor suppressor functional inactivation.
Literature	GENERIF_SUMMARY	The current findings demonstrated that loss of Rb and p16/INKa expression and high E2F1 expression indicate impairment of the Rb suppressor pathway
Literature	GENERIF_SUMMARY	A rare gliosarcoma with liposarcomatous differentiation showed alteration of pRB in sarcomatous component.
Literature	GENERIF_SUMMARY	the Rb/Raf-1 interaction has a role in cell proliferation and angiogenesis
Literature	GENERIF_SUMMARY	We screened for RB1 gene sequence alterations in both peripheral blood and tumor specimens from a total of 48 Mexican retinoblastoma patients...mutation analysis was not helpful to distinguish sporadic and hereditary retinoblastoma
Literature	GENERIF_SUMMARY	We did not observed any correlation between Rb and other clinocopathological features
Literature	GENERIF_SUMMARY	Rb inactivation contributes to tumor progression due to not only loss of cell proliferation control but also conversion to an invasive phenotype.
Literature	GENERIF_SUMMARY	The first in vitro evidence for an important role of C DKN2A/p16 and CDK4 in chondrosarcoma cell survival and proliferation, is presented.
Literature	GENERIF_SUMMARY	SerpinB2 is a cell survival factor that modulates Rb repression of proapoptotic signal transduction
Literature	GENERIF_SUMMARY	Low-penetrant RB allele in small-cell cancer shows geldanamycin instability and discordant expression with mutant ras.
Literature	GENERIF_SUMMARY	Mutations in RB1 in melanoma by inhibition of nonsense-mediated mRNA decay.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	With tumor suppressor protein p53, RB1 is involved in proteasome inhibition cell senescence.
Literature	GENERIF_SUMMARY	These results indicate that PRMT5 overexpression epigenetically alters the transcription of key tumor suppressor genes.
Literature	GENERIF_SUMMARY	phosphatases and pRB have important roles in IGF-I/mTOR-mediated cell survival
Literature	GENERIF_SUMMARY	Karyotyping and FISH assessment for chromosomal abnormalities in the RB1 gene corrates with survival in B-cell chronic lymphocytic leukemiae.
Literature	GENERIF_SUMMARY	MYPT1 may regulate the phosphorylation level of pRb, thereby it may be involved in the control of cell cycle progression and in the mediation of chemoresistance of leukemic cells.
Literature	GENERIF_SUMMARY	The patient was found to carry a de novo heterozygous deletion (g.59444 del196) that results in mis-splicing of exon 8 of RB1, producing a non-functional retinoblastoma protein.
Literature	GENERIF_SUMMARY	The extreme COOH terminus of the retinoblastoma tumor suppressor protein pRb is required for phosphorylation on Thr-373 and activation of E2F.
Literature	GENERIF_SUMMARY	Both siblings in each family described in this report developed unilateral unifocal retinoblastoma as a new germ-line RB1 mutation.
Literature	GENERIF_SUMMARY	Functional loss of RB1 is common in basal-like tumours, which may play a key role in dictating their aggressive biology and unique therapeutic responses.
Literature	GENERIF_SUMMARY	The p16INK4a/cyclin D1/pRB pathway was altered in gastrointestinal tract endocrine tumors, and the loss of expression of pRB may be helpful in identifying patients at high risk of metastasis in rectal well-differentiated endocrine neoplasms
Literature	GENERIF_SUMMARY	by retaining RB1 and amplifying CDK8, colorectal tumour cells select conditions that collectively suppress E2F1 and enhance the activity of beta-catenin
Literature	GENERIF_SUMMARY	Proto-oncogene FBI-1 (Pokemon/ZBTB7A) represses transcription of the tumor suppressor Rb gene via binding competition with Sp1 and recruitment of co-repressors.
Literature	GENERIF_SUMMARY	These results indicate a ROS-independent but p53/retinoblastoma protein-dependent senescence mechanism during hyperoxia.
Literature	GENERIF_SUMMARY	These findings outline the potential role of pRB immunostaining in predicting stage T1 grade 3 bladder cancer response to BCG immunotherapy
Literature	GENERIF_SUMMARY	Expression of cyclin E could not predict survival in patients with gastric cancer, whereas the expression of pRb and p21 was associated with a favorable

Main Category	Category	Terms
		prognosis.
Literature	GENERIF_SUMMARY	Analysis of pRB expression in aggressive fibromatosis (desmoid tumor) is reported.
Literature	GENERIF_SUMMARY	Epigenetic inactivation of the RB1 gene as a factor of genomic instability
Literature	GENERIF_SUMMARY	K-RAS point mutations, and anomalies of p16-RB1-cyclin D pathway could occur before LOH on 10q23 (PTEN) and microsatellite instability during tumor progression.
Literature	GENERIF_SUMMARY	Immunohistochemical analysis of Rb protein expression in neoplastic cells made it easier to evaluate the mechanisms of cancerogenesis in laryngeal cancer and is closely related to genetic alteration in Rb locus.
Literature	GENERIF_SUMMARY	AICAR and glucose deprivation might induce G1/G0 cell cycle arrest and suppress proliferation of NSCs via phospho-Rb and cyclin D down-regulation
Literature	GENERIF_SUMMARY	positive expression of retinoblastoma protein and cyclinD1 might be an early event in gastric carcinoma
Literature	GENERIF_SUMMARY	Reduction in Rb1 expression due to methylation is associated with bladder cancer.
Literature	GENERIF_SUMMARY	Cellular growth inhibition by FK778 is linked to G1 arrest or S phase accumulation, dependent on the functional status of the retinoblastoma protein.
Literature	GENERIF_SUMMARY	Aurora B determines cellular fate after an aberrant mitosis by directly regulating the Rb tumor suppressor protein.
Literature	GENERIF_SUMMARY	Data show that RB is associated with promoters in a manner similar to p107/p130 and that association is modulated by phosphorylation during cell cycle progression.
Literature	GENERIF_SUMMARY	Mosaicism for an RB1 mutation is highly likely to manifest as retinoblastoma.
Literature	GENERIF_SUMMARY	alterations in Rb expression may be essential for both the establishment and maintenance of phenotypic characteristics of Uterine carcinosarcomas
Literature	GENERIF_SUMMARY	identified 17 occurrences of 11 distinct germline mutations in two sporadic and in 15 familial cases of retinoma; the 11 identified mutations were located in exons 1, 10,11,13,14, and 19 to 23; 4 of the identified mutations were not previously reported
Literature	GENERIF_SUMMARY	Data show that E2 induces a generalized re-localization of E7 within the cell nucleus, one potential consequence of which is the inhibition of E7-induced degradation of pRb.
Literature	GENERIF_SUMMARY	Results show that overexpression of CDK4, cyclin D1, and Rb proteins, and loss of expression of proteins p16, p27, and p19 were statistically significant in NPC

Main Category	Category	Terms
		tissues compared with non-cancerous NPE (P
Literature	GENERIF_SUMMARY	Defect in virus replication is resulted from the inability of the mutated NS5B to regulate Rb abundance and thereby modulate host gene expression.
Literature	GENERIF_SUMMARY	the expression of p53, p21 and RB was not significantly different in penile verrucous carcinoma than in usual type squamous cell carcinoma.
Literature	GENERIF_SUMMARY	Cdh1 reciprocally regulates the Rb pathway through competing with E2F1 to bind the hypophosphorylated form of Rb.
Literature	GENERIF_SUMMARY	Patients with Triple-negative breast cancer lacking the retinoblastoma protein had a very favorable clinical outcome if treated with conventional adjuvant chemotherapy.
Literature	GENERIF_SUMMARY	Data show that Bac-ELP1-p21 caused a decrease in Rb phosphorylation levels in cells treated at 42 degrees C.
Literature	GENERIF_SUMMARY	miR-17-92 overexpression may serve as a fine-tuning influence to counterbalance the generation of DNA damage in RB-inactivated SCLC cells.
Literature	GENERIF_SUMMARY	Data demonstrate that the novel anticancer mechanism of hinokitiol involves accumulation of p27, down-regulation of pRb, Skp2, and impairment of Cdk2 function.
Literature	GENERIF_SUMMARY	Data show that sorafenib inhibited the raf/ MEK/ERK signal pathway, and downregulated cyclin D1 and Rb levels.
Literature	GENERIF_SUMMARY	Identification of RB1 mutations in hereditary retinoblastoma provides accurate risk prediction and valuable genetic counseling for patients and their families
Literature	GENERIF_SUMMARY	Data show that nutlin-3 activates p53 resulting in a dramatic increase in MDM2 expression and a marked reduction in total Rb protein levels.
Literature	GENERIF_SUMMARY	Cyclin D1 and pRb showed a progressive increase in expression from benign to malignant tumors but with relative decreases in the more advanced tumors (thick primaries and metastatic melanomas).
Literature	GENERIF_SUMMARY	Data show that the inhibition of cell proliferation with L.lac CF was associated with an increase in p53 and p21 expression, the reduction of cyclin D1 expression, and retinoblastoma protein phosphorylation.
Literature	GENERIF_SUMMARY	Results show a cooperative relationship between the expression of Ras and loss of RB in providing a discrete proliferative advantage under physiologically relevant conditions.
Literature	GENERIF_SUMMARY	Data show that SHP-1 knockdown increases p27stability, decreases the CDK6 levels, inducing retinoblastoma protein hypophosphorylation,

Main Category	Category	Terms
		downregulation of cyclin E and thereby a decrease in the CDK2 activity.
Literature	GENERIF_SUMMARY	The majority of tumours showed strong p16, p21, p27, pRb and cyclin D1 staining and little or no p53 expression. Tumours harbouring dysplasia were significantly more likely to be p53-positive and exhibit up-regulated p21 and p27.
Literature	GENERIF_SUMMARY	TFII-I plays an inhibitory role in regulating genes that are essential in osteogenesis and intersects with the bone-specific transcription factor Runx2 and the retinoblastoma protein, pRb.
Literature	GENERIF_SUMMARY	Retinoblastoma protein-depleted human fibroblasts showed the presence of micronuclei containing whole chromosomes besides the presence of supernumerary centrosomes and aneuploidy.
Literature	GENERIF_SUMMARY	Exposure of human cancer cell lines to caffeine and CGK733 was associated with a reduction in the levels of both phosphorylated and total retinoblastoma protein (
Literature	GENERIF_SUMMARY	44% of Paget's disease of the vulva cases examined showed loss of expression of RB protein compared with 67% of Paget's disease of the breast cases.
Literature	GENERIF_SUMMARY	RB is a direct target of miR-675 which suppressed a reporter carrying the RB 3'-UTR containing the miR-675-binding site. Transfection with anti-miR-675 increased RB expression & decreased cell growth & soft agar colony formation in colon cancer cells.
Literature	GENERIF_SUMMARY	findings indicate a novel aspect of pRb acute loss that promotes aneuploidy mainly by inducing CENPA overexpression
Literature	GENERIF_SUMMARY	There are specific interactions among Rb, HDAC1 and E2F1 proteins in gallbladder carcinoma, indicating the existence of the direct Rb/E2F1/HDAC1 signal transduction pathway. p16 does not interact with any of these proteins.
Literature	GENERIF_SUMMARY	RB1 is imprinted in the same direction as CDKN1C, which operates upstream of RB1.
Literature	GENERIF_SUMMARY	the bilaterally affected son and the unilaterally affected father were both heterozygous for the nonsense mutation c.1363C>T (p.R455X) in exon 14 of RB1
Literature	GENERIF_SUMMARY	simultaneous detection of p53, Rb, p16, and EGFR in a suspension microarray facilitates rapid diagnosis of lung cancer
Literature	GENERIF_SUMMARY	Studies developed a combinatorial approach for detection of recurrent mutations, large deletions/ insertions and point mutations of RB1 gene in patients with Rb.
Literature	GENERIF_SUMMARY	CDK4/6 inhibition is a potent mediator of cytostasis and that RB loss can be readily compensated for in the

Main Category	Category	Terms
		context of both hepatoma cell lines and liver tissue.
Literature	GENERIF_SUMMARY	Lysine methylation regulates the pRb tumour suppressor protein.
Literature	GENERIF_SUMMARY	Results suggest that reciprocal Notch signaling might be necessary for the proliferation and survival of HL60 cells, possibly through the maintenance of the expression of c-Myc and Bcl2, as well as the phosphorylation of the Rb protein.
Literature	GENERIF_SUMMARY	Phosphorylation-induced conformational changes in the retinoblastoma protein inhibit E2F transactivation domain binding.
Literature	GENERIF_SUMMARY	LIMD1 inactivation as primary event than inactivation of RB1 in head and neck squamous cell carcinoma development.
Literature	GENERIF_SUMMARY	Microarray expression analysis indicated that E2F1 and E2F2 are downstream genes in the KDM5B pathway.
Literature	GENERIF_SUMMARY	In the absence of Rb, E2F1 overexpression results in apoptosis, not proliferation, and that this effect is independent of p53.
Literature	GENERIF_SUMMARY	The protein expression profiles of p53, p73, pRb family proteins, and PCNA in different cell lines of mouse primitive neuroectodermal tumors (PNET), and in human medulloblastoma cell lines, was examined.
Literature	GENERIF_SUMMARY	this study more generally suggests an interplay between RB and IDE within the proteasome that may have important growth-regulatory consequences.
Literature	GENERIF_SUMMARY	high telomerase activity with intact p16/Rb could be a marker of stem cell origin in cancer cells
Literature	GENERIF_SUMMARY	stimulation of PP1 activity via siRNA mediated knockdown of its interacting protein PNUTS (Phosphatase Nuclear Targeting Subunit) leads to Rb dephosphorylation and apoptosis in cancer cells
Literature	GENERIF_SUMMARY	two spontaneously transformed human oral keratinocyte lines from adult human gingival samples revealed a mutation in the p53 gene, loss of p21WAF1/Cip1 and overexpression of p-Rb-Ser807
Literature	GENERIF_SUMMARY	Genetic analysis shows mutations of RB1 in all cases of retinoblastoma.
Literature	GENERIF_SUMMARY	analysis of immortalization of oral keratinocytes by functional inactivation of the p53 and pRb pathways
Literature	GENERIF_SUMMARY	in HPV positive oropharyngeal cancer, cyclin D1 has a role in locoregional recurrence, event and death and pRb has a role in event and death
Literature	GENERIF_SUMMARY	Rb1 is an important determinant of Tbx2 functional specificity.
Literature	GENERIF_SUMMARY	pRB depletion compromises centromeric localization of

Main Category	Category	Terms
		CAP-D3/condensin II and chromosome cohesion, leading to an increase in intercentromeric distance and deformation of centromeric structure
Literature	GENERIF_SUMMARY	Findings suggest that Gadd45b mediates p38-induced Rb phosphorylation by enhancing the interaction between p38 and Rb during Fas-induced apoptosis in murine hepatocytes.
Literature	GENERIF_SUMMARY	results show that in RB patients in Malaysia mutation frequency in pocket B of the E1A binding domain of the RB1 gene is lower; mutations in exons outside the pocket B domain may be associated with the disease in Malaysian RB patients
Literature	GENERIF_SUMMARY	findings suggest RB1 mutations to be of pathological importance potentially affecting sensitivity to mitomycin/anthracycline treatment in breast cancer
Literature	GENERIF_SUMMARY	The structural and functional effect of pRB by significant nsSNP in the coding region of RB1 gene is reported.
Literature	GENERIF_SUMMARY	present study indicates that RB1CC1 together with hSNF5 and/or p53 enhances the RB1 pathway through transcriptional activation of RB1, p16 and p21
Literature	GENERIF_SUMMARY	Assaying tumor-specific RB1 mutations complements cytological and immunohistochemical assessment of retinoblastoma involvement of cerebrospinal fluid and bone marrow.
Literature	GENERIF_SUMMARY	Expression of constitutively active mTOR conferred resistance to cryptotanshinone inhibition of cyclin D1 expression and Rb phosphorylation, as well as cell growth.
Literature	GENERIF_SUMMARY	discovery of the first alpha-N-methyltransferase, which we named N-terminal RCC1 methyltransferase (NRMT).
Literature	GENERIF_SUMMARY	E2F-1 associated pRB is modified by O-GlcNAc.
Literature	GENERIF_SUMMARY	Reduced expression of RB is associated with leiomyosarcoma.
Literature	GENERIF_SUMMARY	protein phosphatase 1 competition with Cdk-cyclins for Rb binding is sufficient to retain Rb activity and block cell-cycle advancement.
Literature	GENERIF_SUMMARY	these findings provide the first demonstration that SM22alpha modulates cellular senescence caused by damaging agents via regulation of the p16(INK4a)/pRB pathway in HepG2 cells and that these effects of SM22alpha are partially mediated by MT-1G.
Literature	GENERIF_SUMMARY	RB.E2F1 complex plays a critical role in ZBRK1 transcriptional repression, and loss of this repression may contribute to cellular sensitivity of DNA damage, ultimately leading to carcinogenesis.
Literature	GENERIF_SUMMARY	Data indicate that local Sca-1low dermal mesenchymal stem/progenitor cells are preferential targets for

Main Category	Category	Terms
		malignant transformation associated with deficiencies in both p53 and Rb.
Literature	GENERIF_SUMMARY	RB monomethylation at lysine 860 by SMYD2 provides a direct binding site for L3MBTL1.
Literature	GENERIF_SUMMARY	Data show that phosphorylation by p38 triggers an interaction between Rb and the Hdm2, leading to degradation of Rb and cell death.
Literature	GENERIF_SUMMARY	Altered pRb was significantly correlated with overall survival and the expression pattern of pRb and p53, p53/pRb (altered/normal) had worse survival than p53/pRb (normal/altered).
Literature	GENERIF_SUMMARY	Data show that proliferaton and nuclear expression of cyclin D1 and p-Rb coincide with the early phase of cyst growth, suggesting that there might be a therapeutic window in which cyclin-dependent kinase inhibitors are most effective in ARPKD.
Literature	GENERIF_SUMMARY	acetylation is required for nuclear localization and keratinocyte differentiation
Literature	GENERIF_SUMMARY	These results indicate that the RB-loss signature expression is associated with poor outcome in breast cancer, but predicts improved response to chemotherapy based on data in ER-negative populations.
Literature	GENERIF_SUMMARY	Data provide evidence for a functional link between Hec1 expression and the pRb pathway.
Literature	GENERIF_SUMMARY	Data indicate that miR-449a is a miRNA component of the Rb pathway and its tumor suppressor-like effects, in part, depends on Rb status in prostate cancer cells.
Literature	GENERIF_SUMMARY	CEBPD reverses RB/E2F1-mediated gene repression and participates in HMDB-induced apoptosis of cancer cells.
Literature	GENERIF_SUMMARY	Further investigation is required to assess whether the expression of Rb family proteins can potentially be used as a prognostic or predictive factor in gastric cancer.
Literature	GENERIF_SUMMARY	The methylation status of RB gene was correlated with lymph node metastasis, but not with the size, the differentiation status, ER expression of the breast cancer and the age of the patients.
Literature	GENERIF_SUMMARY	Data show that Orc1 specifically binds hypo-phosphorylated Rb and that this interaction is competitive with the binding of Rb to E2F1.
Literature	GENERIF_SUMMARY	RB controlled nuclear receptor networks critical for tumor progression and that it did so via E2F transcription factor 1-mediated regulation of androgen receptor (AR) expression and output.
Literature	GENERIF_SUMMARY	A new function for these factors in the control of insulin secretion and open up new avenues for the treatment of diabetes.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	In pRb, lysine (K) 810 represents the essential and conserved basic residue (SPXK) required for cyclin/Cdk recognition and phosphorylation.
Literature	GENERIF_SUMMARY	CDK-mediated phosphorylation of both RBP1 and pRb induces their dissociation to mediate release of the mSin3.HDAC transcriptional repressor complex from pRb to alleviate transcriptional repression of E2F.
Literature	GENERIF_SUMMARY	RB1CC1, RB1 and p53 have prognostic roles in breast cancer in Japanese patients
Literature	GENERIF_SUMMARY	Data show that the binding of scFv 43M2 to E7 was inhibited by pRb in a non-competitive manner.
Literature	GENERIF_SUMMARY	RB1 alterations do not play a major role in metastatic development of breast cancer.
Literature	GENERIF_SUMMARY	p16 and retinoblastoma determines response to CDK4/6 inhibition in ovarian cancer
Literature	GENERIF_SUMMARY	Rb-loss sensitizes cells to autophagy via direct and indirect mechanisms.
Literature	GENERIF_SUMMARY	over-expression of miR-132 and miR-212 result in reduced pRb protein in pancreatic cancer cells.
Literature	GENERIF_SUMMARY	knockout of the two alleles of the RB1 gene is necessary and may be largely sufficient for the development of RB, in support of Knudson's two-hit hypothesis.
Literature	GENERIF_SUMMARY	Retinoblastoma protein controls the human CD34 promoter region by antagonizing the CD34 promoter factor nucleolin to provide a mechanism that links expression of endogenous CD34 to cell cycle progression.
Literature	GENERIF_SUMMARY	Data show that the expression of CCND1/CDKN2A correlated with RB1 status in xenograft tumors in vivo.
Literature	GENERIF_SUMMARY	a fraction of low-grade diffuse gliomas lacking common genetic alterations may develop through a distinct genetic pathway, which may include loss of cell-cycle control regulated by the RB1 pathway.
Literature	GENERIF_SUMMARY	ARF1 controls proliferation of breast cancer cells by regulating the retinoblastoma protein
Literature	GENERIF_SUMMARY	The study found conditions in which crystals of the retinoblastoma protein pocket domain could be grown in the absence of any ligand, and we solved the structure to 2.5 A resolution.
Literature	GENERIF_SUMMARY	Cancerous tissues from 30 patients with HPSCC were examined for LOH in 4 tumor suppressor genes (TSGs) (p16, Rb, E-cadherin, and p53) at loci 9p21, 13q21, 6q22, and 17p13, respectively, using microsatellite markers amplified by polymerase chain reaction.
Literature	GENERIF_SUMMARY	Results reveal a functional connection between the pRB and Hippo tumor suppressor pathways, and suggest

Main Category	Category	Terms
		that low levels of LATS2 may undermine the ability of pRB to induce a permanent cell cycle arrest in tumor cells.
Literature	GENERIF_SUMMARY	p16INK4a/pRb may function as an alternative pathway of oncogene-induced senescence; the reactivation of p16INK4a expression might be a novel strategy to restore the senescence program in some tumors.
Literature	GENERIF_SUMMARY	C/EBPbeta reduces monocytic proliferation by affecting the retinoblastoma/E2F/cyclin E pathway and that it may contribute to, but is not directly required for, macrophage morphology.
Literature	GENERIF_SUMMARY	Single nucleotide polymorphisms in p16, RB1 genes may affect the susceptibility to cervical cancer collectively.
Literature	GENERIF_SUMMARY	these findings are consistent with a causative link between defects in the BRCA1 gene by CpG island hypermethylation and the development of TNBCs characterized by RB/p16 dysfunction in sporadic cases of the disease.
Literature	GENERIF_SUMMARY	results suggest that, in cancer, frequent and persistent expression of PAX8 is required for cell growth control through transcriptional activation of E2F1 expression and upregulation of the RB-E2F1 pathway
Literature	GENERIF_SUMMARY	screened 30 family members from a retinoblastoma family;phenotypic variability associated with p.V654L mutation in exon 19 of RB1 gene in 11 family members was found; suggests RB1 p.V654L mutation is a typical mutation associated with low penetrance
Literature	GENERIF_SUMMARY	study included 5 families with more than 1 member affected with retinoblastoma; study reveals novel effects of RB1 mutations on splicing
Literature	GENERIF_SUMMARY	CTCF plays an important role in maintaining the Rb promoter in an optimal chromatin configuration.
Literature	GENERIF_SUMMARY	show that chromosome instability (CIN) in cultured cells lacking Rb family proteins requires Mad2 upregulation and that this upregulation is also necessary for CIN and tumor progression in vivo
Literature	GENERIF_SUMMARY	Linkage analysis revealed inheritance of RB1 alleles in 10 families with retinoblastoma.
Literature	GENERIF_SUMMARY	identified concurrent mutational inactivation of the PTEN and RB1 tumor suppressors as a mechanism for loss of BRAF/MEK dependence in melanomas harboring (V600E)BRAF mutations
Literature	GENERIF_SUMMARY	downstream effectors of the p53-p21 and p16-pRB tumour suppressor pathways crucial for mediating entry into senescence
Literature	GENERIF_SUMMARY	Integrin subunits alpha5 and alpha6 regulate cell cycle by modulating the chk1 and Rb/E2F pathways to affect breast cancer metastasis.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Eight known mutations of RB1 and four novel mutations (c.832_833insT, c.1943delC, c.1206C>T, and c.2029delG) were determined
Literature	GENERIF_SUMMARY	suggests that the pRb protein level is a promising marker for polyamine depletion sensitivity and that there is a connection between pRb and the regulation of SSAT activity
Literature	GENERIF_SUMMARY	Bcl-2 gene transcription is regulated by Ap2alpha-Rb complex binding to the Bcl-2 promoter.
Literature	GENERIF_SUMMARY	The LXCXE binding motif has an essential role in RB-mediated transcriptional control, response to genotoxic insult, and tumor suppression.
Literature	GENERIF_SUMMARY	Data suggest that the RB pathway may prevent hepatocellular carcinoma development by maintaining the quiescence of adult liver progenitor cells.
Literature	GENERIF_SUMMARY	the role of Rb in mediating the inverse relationship between elastogenesis and cellular proliferation.
Literature	GENERIF_SUMMARY	results suggest that in malignant epithelial cells beta-defensin-2 (BD-2)may control cell growth via arrest of G1/S transition and activation of retinoblastoma protein
Literature	GENERIF_SUMMARY	genetics can be incorporated into the management plan of all retinoblastoma patients using a team approach to ensure timely evaluations and appropriate counseling.
Literature	GENERIF_SUMMARY	An association between IVF or ICSI and retinoblastoma through this epigenetic mechanism is unlikely.
Literature	GENERIF_SUMMARY	Complete loss of RB1 transcript expression is seen in 18 of 170 glioblastoma tumors which were highly enriched for, but not exclusive to, the proneural subtype.
Literature	GENERIF_SUMMARY	Rb resulted in the c-Myc up-regulation in the liver of mt-Rb transgenic mice and then c-Myc activated several target genes in the tumor.
Literature	GENERIF_SUMMARY	DNA damage signals alter pRB and E2F1 to engage them in functions leading to apoptotic induction that are distinct from pRB-E2F regulation in cell cycle control
Literature	GENERIF_SUMMARY	study suggests a weak paternal age effect for retinoblastom resulting from de novo germline mutations consistent with the paternal origin of most of these mutations
Literature	GENERIF_SUMMARY	RB1 genotype-phenotype correlation for second primary cancers of the hereditary retinoblastoma survivors
Literature	GENERIF_SUMMARY	Ten RB1 germline mutations have been identified in 41 unrelated Moroccan patients with retinoblastoma.
Literature	GENERIF_SUMMARY	RB inactivation contributes to tumor progression not only through loss of cell cycle control but also through up-regulation of ZEB expression and induction of an invasive phenotype.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	The spectrum of germline and tumoral alterations included
Literature	GENERIF_SUMMARY	Studies indicate that the Ser/Thr phosphatases PP1 and PP2A are responsible for the dephosphorylation and activation of Rb proteins.
Literature	GENERIF_SUMMARY	E2F1 suppresses TLR3 transcription, but during immune stimulation, Rb is upregulated to block the inhibitory effect of E2F1 on TLR3, highlighting a role of Rb-E2F1 axis in the innate immune response in epithelial cells
Literature	GENERIF_SUMMARY	The pRb isomerization done by Pin1 is essential for further phosphorylation of pRb and consequent inactivation.
Literature	GENERIF_SUMMARY	The individual RB family proteins were sufficient for the establishment of p16-initiated senescence, and that growth arrest in G 1 was not dependent on the presence of functional pRb or p53.
Literature	GENERIF_SUMMARY	Data show that expression of RB1 repressed both the wild-type and let-7-mutated promoter.
Literature	GENERIF_SUMMARY	Studies indicate that retinoblastoma (RB) protein might possess many multifaceted and unique functions.
Literature	GENERIF_SUMMARY	in various tumors types, there is an inverse correlation between the expression levels of gamma-tubulin and RB1 and that in tumor cell lines with a nonfunctioning RB1, reduction of gamma-tubulin protein levels leads to induction of apoptosis.
Literature	GENERIF_SUMMARY	Study reveals a novel role for Rb protein N-terminal in the mechanism of Rb inactivation and provides the first insights into the overall structure of the multidomain Rb protein. The distinct structural changes induced by particular phosphorylation events explain how multisite phosphorylation can differentially regulate Rb interactions with other proteins.
Literature	GENERIF_SUMMARY	the paramount expression of these cellular cycle biomarkers (p53 and Rb protein) in excisional biopsies may suggest that molecular lesion assessment plays a role in pre-malignant lesions.
Literature	GENERIF_SUMMARY	All cases of cellular angiofibroma, spindle cell lipoma and mammary-type myofibroblastoma showed monoallelic or biallelic loss of RB1. The absence of RB1 loss in solitary fibrous tumors suggests that they are not related.
Literature	GENERIF_SUMMARY	Inactivation of Rb in stromal fibroblasts promotes epithelial cell invasion.
Literature	GENERIF_SUMMARY	This report identifies two additional PTPs, SHP-2 and PTP-PEST, that are also directly activated by the Rb-E2F pathway and which can contribute to signal transduction during p53-independent apoptosis.
Literature	GENERIF_SUMMARY	Loss of heterozygosity (LOH) of the retinoblastoma

Main Category	Category	Terms
		gene was detected in 25.6% of patients in intron 1 and in 37.2% of patients in intron 17.
Literature	GENERIF_SUMMARY	Although the retinoblastoma protein (Rb) is known to regulate the growth of keratinocytes in a cell-autonomous manner, here we describe a function of Rb in the stromal compartment.
Literature	GENERIF_SUMMARY	Data indicate that endoreduplication and mitotic failure occur during telomere crisis in fibroblasts and mammary epithelial cells, and suggest the role of p53 and Rb in repressing tetraploidization.
Literature	GENERIF_SUMMARY	Rb1 induction in chronic asymptomatic HIV-1 infection mediates apoptosis resistance in monocytes in association with protective autophagy and contributes to monocyte survival during immune activation and/or HIV-1 viremia.
Literature	GENERIF_SUMMARY	RB1 was identified as a major target for genomic disruption in tumors arising in BRCA1 mutation carriers and in sporadic tumors with BRCA1 promoter methylation but rarely in other breast cancers.
Literature	GENERIF_SUMMARY	study observed associations between father's diet before his child's conception and risk of retinoblastoma in his child that results from a new germline RB1 mutation
Literature	GENERIF_SUMMARY	Rb1 germline mutations may contribute to genetic susceptibility for head and neck cancer.
Literature	GENERIF_SUMMARY	-dependent RB1 methylation at lysine 810 promotes cell cycle progression of cancer cells. Further study may explore SMYD2-dependent RB1 methylation as a potential therapeutic target in human cancer.
Literature	GENERIF_SUMMARY	of the soft tissue tumors associated with 13q deletions, all spindle cell lipomas, pleomorphic lipomas, and cellular angiofibromas and most mammary-type myofibroblastomas show loss of Rb expression
Literature	GENERIF_SUMMARY	identify a causal role for E2F7 in cellular senescence and uncover a novel link between the RB and p53 pathways.
Literature	GENERIF_SUMMARY	RB1 deletions are associated with relapse in multiple myeloma.
Literature	GENERIF_SUMMARY	Dephosphorylation of threonine-821 of the retinoblastoma tumor suppressor protein (Rb) is required for apoptosis induced by UV and Cdk inhibition.
Literature	GENERIF_SUMMARY	Novel molecular mechanisms for the different expression patterns of E2F1 targets and the selectivity of retinoblastoma protein-mediated heterochromatin formation and gene repression during cellular senescence.
Literature	GENERIF_SUMMARY	pRB plays important roles through multiple functions in determining cell fate, i.e., normal growth/death and

Main Category	Category	Terms
		differentiation, or tumor formation.
Literature	GENERIF_SUMMARY	the Rb gene regulatory network (GRN) subcircuit that regulates cell proliferation is linked to a GRN subcircuit regulating a tumor cell immune function.
Literature	GENERIF_SUMMARY	A novel function for Rb is promoting cell survival through regulating the function of BAG-1.
Literature	GENERIF_SUMMARY	Rb1 is a direct functional target of miR-26a.
Literature	GENERIF_SUMMARY	Molecular modeling and MD simulations of pRB in free and ligand-bound forms confirms that pRB could be a valid target for the anticancer drug design when the cancer is induced by the viral onco-proteins and forms a clear base for designing E7 antagonists.
Literature	GENERIF_SUMMARY	results suggest that miR-106a might have a regulatory role for Rb1 in sporadic colorectal cancer
Literature	GENERIF_SUMMARY	High-grade bladder in situ/muscle-invasive carcinoma is characterized by alteration of p53 and pRB.
Literature	GENERIF_SUMMARY	Clinicians should consider retinoblastoma as a diagnosis when faced with a patient with unexplained exudative retinal detachment.
Literature	GENERIF_SUMMARY	This study enabled us to obtain information about molecular and genetic features of patients with retinoblastoma in Argentina and correlate them to their phenotype.
Literature	GENERIF_SUMMARY	our data identify a potential role for pRB in regulating angiogenesis through modulation of Nuclear protein 7 expression.
Literature	GENERIF_SUMMARY	Rb1 seems to play a permissive role for human adipose tissue function, being downregulated in obesity and increased during differentiation of human adipocytes.
Literature	GENERIF_SUMMARY	these results suggest that nuclear beta-catenin-dependent RB stabilization suppresses TNF-induced apoptosis in caspase-8-positive colon cancer cells.
Literature	GENERIF_SUMMARY	This review emphasizes the importance of the mechanisms of RB1 gene inactivation in inducing cancer cell development.
Literature	GENERIF_SUMMARY	Silencing of RB1 but not RB2/P130 decreases proliferative activity and impairs the differentiation potential of human mesenchymal stem cells.
Literature	GENERIF_SUMMARY	Loss of function of pRB was associated with tumor invasion and lymph node metastasis in esophageal squamous cell carcinoma. [Meta-Analysis]
Literature	GENERIF_SUMMARY	Authors show that only the proteasome 19S regulatory particle that normally participates in ubiquitin-dependent protein degradation, is required for human herpesvirus 5 pp71-mediated degradation of Rb and Daxx.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	pRb inactivation is insufficient to restore efficient viral nuclear egress of human cytomegalovirus in the absence of UL97.
Literature	GENERIF_SUMMARY	caspases contribute to trigger the mitochondrial apoptotic pathway by coupling Bcl-2/Bcl-xL inhibition to that of Mcl-1, via the pRb/E2F-1-dependent induction of Noxa
Literature	GENERIF_SUMMARY	Mutations in the RB1 gene are screened in a large cohort of Iranian patients using the multiplex ligation-dependent probe amplification (MLPA) technique for detecting deletions/duplications in retinoblastoma.
Literature	GENERIF_SUMMARY	p53 promotes VEGF expression and angiogenesis in the absence of an intact p21-Rb pathway.
Literature	GENERIF_SUMMARY	Amplification of the MYCN oncogene might initiate retinoblastoma in the presence of non-mutated RB1 genes. These unilateral RB1(+/+)MYCN(A) retinoblastomas are characterised by distinct histological features.
Literature	GENERIF_SUMMARY	Fifteen substitutions (4 intronic and 11 exonic) were identified as probably or likely pathogenic. Four of these 11 exonic substitutions were novel.
Literature	GENERIF_SUMMARY	Mutation in the RB1 gene predisposes to heritable retinoblastoma and to further secondary primary malignancies. Abnormal Rb protein may play a role in the pathogenesis of calcific aortic valve disease in the context of familial bicuspid aortic valves.
Literature	GENERIF_SUMMARY	pRB potentiated TNFalpha-induced apoptosis even when translation was blocked.
Literature	GENERIF_SUMMARY	The Rb1 gene inhibits the viability of retinoblastoma cells by regulating homologous recombination.
Literature	GENERIF_SUMMARY	EZH2-mediated signal loop, EZH2-CDK4/6-pRb-E2F1, is probably involved in GBM tumorigenicity, and this loop can be blocked by miRNA-138
Literature	GENERIF_SUMMARY	3 new mutations were found in spontaneous retinoblastoma
Literature	GENERIF_SUMMARY	the cooperative effect of inactivation of both pRB and p53 tumor suppressor pathways promotes Whole chromosome instability
Literature	GENERIF_SUMMARY	The present study failed to demonstrate any conclusive role of cyclin D1/RB/ p16 pathway in carcinoma of the gallbladder.
Literature	GENERIF_SUMMARY	Data indicate there was no statistical significance observed between the immunoexpression of p16, p21, p27, and pRb with nodal metastasis, depth of invasion, and presence of vascular or perineural invasion.
Literature	GENERIF_SUMMARY	Late events such as the loss of RB1 and gain of MDM2 appear to be related to the final mechanisms contributing to the metastatic establishment of

Main Category	Category	Terms
		osteosarcoma.
Literature	GENERIF_SUMMARY	MAP kinase p38 and retinoblastoma protein signaling is required for DNA damage-mediated formation of senescence-associated heterochromatic foci in tumor cells.
Literature	GENERIF_SUMMARY	We investigated the association between Rb and p16 protein expression and clinical outcome in no-special-type triple-negative breast carcinomas.
Literature	GENERIF_SUMMARY	CASZ1 inhibits cell cycle progression in neuroblastoma by restoring retinoblastoma protein activity.
Literature	GENERIF_SUMMARY	deletions of RB1 and CDKN2A genes may contribute to the development of hyperleukocytic type of T-lineage acute lymphoblastic leukemia in children
Literature	GENERIF_SUMMARY	This study suggests the alteration of p16 and pRb expression has been detected in carcinoma ex pleomorphic adenomas
Literature	GENERIF_SUMMARY	50% of malignant phyllodes display evidence of Rb/p16 pathway alterations, likely reflecting p16 or Rb inactivation.
Literature	GENERIF_SUMMARY	Involvement of SNIP1 in progression of lung cancer by regulating the RB/HDAC1 interaction.
Literature	GENERIF_SUMMARY	WT1 is an oncogene and promotes NSCLC cell proliferation by up-regulating Cyclin D1 and p-pRb expression
Literature	GENERIF_SUMMARY	RB represents a molecular link between autophagy and apoptosis, and a resistance marker in glioblastomas.
Literature	GENERIF_SUMMARY	RB phosphorylation and beta-catenin nuclear translocation are differently expressed in Crohn's disease and ulcerative colitis
Literature	GENERIF_SUMMARY	Genomic deregulation of the E2F/Rb pathway leads to activation of the oncogene EZH2 in small cell lung cancer.
Literature	GENERIF_SUMMARY	Rb protein is essential to the senescence-associated heterochromatic foci formation induced by HMGA2.
Literature	GENERIF_SUMMARY	miR-215 targets retinoblastoma tumor-suppressor gene 1 (RB1) through its 3'-UTR in gastric cancer cells.
Literature	GENERIF_SUMMARY	RB1 recurrent mutations in retinoblastoma analized by ARMS-PCR.
Literature	GENERIF_SUMMARY	Expression of cell cycle regulatory factors hus1, gadd45a, rb1, cdkn2a and mre11a correlates with expression of clock gene per2 in human colorectal carcinoma tissue.
Literature	GENERIF_SUMMARY	Sixteen germline mutations were identified in Chinese patients with retinoblastoma.
Literature	GENERIF_SUMMARY	Authors find that phosphorylation of S788 and S795

Main Category	Category	Terms
		weakens the direct association between the N-terminal portion of the retinoblastoma protein C-terminal domain and the marked-box domains of E2F.
Literature	GENERIF_SUMMARY	A key role for the RB pathway in invasion associated with breast tumor progression.
Literature	GENERIF_SUMMARY	Cdk5 signaling via Rb is critical to medullary thyroid carcinoma tumorigenesis and progression
Literature	GENERIF_SUMMARY	Nimbolide mediated inhibition of retinoblastoma protein phosphorylation causes cell-cycle arrest and suppresses glioblastoma growth.
Literature	GENERIF_SUMMARY	PRMT5 knockdown in non-Hodgkin lymphoma cell lines and primary lymphoma cells leads to RBL2 derepression and RB1 reactivation, which in turn inhibit PRC2 expression.
Literature	GENERIF_SUMMARY	Studied the spectrum of genetic and epigenetic changes of RB1 identified in 194 Retinoblastoma tumours and 209 blood samples, from 403 unrelated RB patients.
Literature	GENERIF_SUMMARY	RB1 expression correlated with that of the senescence-associated secretory phenotype cassette in human osteosarcomas. Low expression of both RB1 & the SASP genes was associated with poor prognosis.
Literature	GENERIF_SUMMARY	CDK2 knockdown alters the profile of Rb phosphorylation in coronary artery smooth muscle cells, as well as the proliferative response of these cells to mitogenic stimulation.
Literature	GENERIF_SUMMARY	high rate/frequency of p14ARF, p15INK4B, and p16INK4A methylation was detected in hepatocellular carcinoma (HCC) with hepatitis B virus infection than in HCC without virus infection
Literature	GENERIF_SUMMARY	RB1 mutations were sought in retinoblastoma patients who also had pineal cysts. Germline mutations were found in 100% of bilateral retinoblastomas and 25% of unilateral ones. Nonsense-frameshift mutations were more common in bilateral disease.
Literature	GENERIF_SUMMARY	the inactivation of RB1 or RB2/P130 in uncommitted bone marrow stromal cells facilitates the first steps of adipogenesis.
Literature	GENERIF_SUMMARY	Deletion of Rb in renin-expressing compartment of pancreas leads to highly penetrant metastatic pancreatic neuroendocrine carcinoma.
Literature	GENERIF_SUMMARY	results reveal a novel non-nuclear function for Rb and suggest a potential mechanism of TNF-alpha-induced disruption of sarcomeric organization
Literature	GENERIF_SUMMARY	Data indicate that seven genes showed statistical enrichment for mutation
Literature	GENERIF_SUMMARY	Rb gene loss is a critical event in the development of small cell carcinomas.

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Literature	GENERIF_SUMMARY	Rb suppresses collective invasion, circulation and metastasis of breast cancer cells in CD44-dependent manner.
Literature	GENERIF_SUMMARY	Retinoblastoma dysfunction converts TGF-beta to a mitogen that activates known oncogenic signaling pathways and upregulates Wnt7b.
Literature	GENERIF_SUMMARY	E2F1/Rb pathways are altered by HHV-6, which then uses the E2F1 transcription factor to express viral genes
Literature	GENERIF_SUMMARY	Analyses of Rb pathway mechanistic differences revealed frequent RB1 genomic alterations such as RB1 allelic loss and/or large spanning deletions in high-grade serous carcinoma.
Literature	GENERIF_SUMMARY	Parallel analysis of somatic mutations among known cancer genes allowed us to identify RB1 as a new actor in aggressive adrenocortical carcinomas.
Literature	GENERIF_SUMMARY	Retinoblastoma protein (pRB) have a novel function in regulating the mitotic function of NuMA and spindle organization, which are required for proper cell cycle progression.
Literature	GENERIF_SUMMARY	LT-Ag bind to Rb and release Rb-dependent repression of E2F-regulated transcription.
Literature	GENERIF_SUMMARY	Retinoblastoma pathway dysregulation leads to an overlapping Merkel cell carcinoma phenotype and oncogenesis occurs through either a polyomavirus-dependent (viral large T antigen expression) or polyomavirus-independent (host somatic mutation) mechanism.
Literature	GENERIF_SUMMARY	This is the first report of a 13q deletion associated with infantile hemangioendothelioma of the liver.
Literature	GENERIF_SUMMARY	Provide evidence of posttranscriptional downregulation of Smurf2 in triple-negative breast cancers, and demonstrate that the loss of RB function is involved in microRNA-mediated interference with Smurf2 translation.
Literature	GENERIF_SUMMARY	This is the first report of chromothripsis as a mechanism for RB1 gene inactivation in cancer.
Literature	GENERIF_SUMMARY	Enhancement of chromosome cohesion suppresses genome instability in pRB-deficient cells.
Literature	GENERIF_SUMMARY	Studied RB1 mutations in a unique nationwide cohort of Rb patients from the Netherlands.
Literature	GENERIF_SUMMARY	Data show that the crosstalk between the Rb-AKT-signaling pathways forms this switch by controlling the overlapping functions of FoxO3a and FoxM1 transcription factors in cultured fibroblasts.
Literature	GENERIF_SUMMARY	Together these findings identify a novel physical and functional interaction between EAF2 and the Rb pathway.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	studies revealed upregulation of p16, pRb dephosphorylation and its interaction with histone deacetylase 1/2
Literature	GENERIF_SUMMARY	Our findings suggest that EZH2-specific microRNA-98 can effectively inhibit cell proliferation in vitro and regulate the pRb-E2F pathway in human ovarian cancer
Literature	GENERIF_SUMMARY	Thirty-four distinct mutations were identified in RB1 gene in 40 (47.1%) of the 85 probands (36 bilateral and four unilateral), of which 20% (8/40) was identified by multiplex ligation-dependent probe amplification.
Literature	GENERIF_SUMMARY	our results demonstrate a combined mechanism for the anti-proliferative effects of FRK by inhibiting cyclin D1 nucleus accumulation and pRb phosphorylation in glioma cells.
Literature	GENERIF_SUMMARY	mutations can be used as a prognostic molecular biomarker for resectable hepatocellular carcinoma
Literature	GENERIF_SUMMARY	PTP-1B, SHP-2, and PTEN all play a functional role in Rb/E2F-associated apoptotic signal transduction
Literature	GENERIF_SUMMARY	Identification of RB1 gene alterations in patients with retinoblastoma improves clinical management of patient and relatives at risk
Literature	GENERIF_SUMMARY	MDM2 serves as a better target for HAUSP-mediated deubiquitination in cancer cells, facilitating degradation of Rb and oncogenic progression.
Literature	GENERIF_SUMMARY	Retinoblastoma protein potentiates the innate immune response in hepatocytes.
Literature	GENERIF_SUMMARY	Loss of imprinting represents an important additional mechanism for RB1 pathway inactivation in human hepatocellular carcinoma, complementing well-described molecular defects.
Literature	GENERIF_SUMMARY	retinoblastoma RB1 -/- cells cannot survive in the absence of MED4, both in vitro and in orthotopic xenograft models in vivo, therefore identifying MED4 as a survival gene in retinoblastoma.
Literature	GENERIF_SUMMARY	Cyclin D has a role in activating the Rb tumor suppressor by mono-phosphorylation
Literature	GENERIF_SUMMARY	Findings indicate that both genetic and epigenetic retinoblastoma gene (RB1) changes may contribute to the pathogenesis of head and neck cancer (HNC) in the Pakistani population.
Literature	GENERIF_SUMMARY	The girl with diffuse infiltrating RB was found to be heterozygous for an oncogenic mutation in the RB1 gene that was also carried by both brothers and the father of the family
Literature	GENERIF_SUMMARY	Frequent disruption of the RB pathway in indolent follicular lymphoma suggests a new combination therapy.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Rb is present in HIF1alpha-ARNT/HIF1beta transcriptional complexes associated with TRIP230 as determined by co-immuno-precipitation, GST-pull-down and ChIP assays
Literature	GENERIF_SUMMARY	the use of p53 and pRB protein expression has no contribution in the situation where there is a diagnostic difficulty in deciding between hepatocellular carcinoma and hepatoblastoma.
Literature	GENERIF_SUMMARY	results showed for the first time that the Rb signaling pathway is essential for mediating the antineoplastic activity of erufosine and its efficacy in patients with malignant diseases may be predicted by determining the Rb status.
Literature	GENERIF_SUMMARY	Ki-67 may be a useful marker for discriminating benign from borderline PTs, and p16 and pRb may be a useful combination of markers for distinguishing FA from benign PTs in core biopsy specimen.
Literature	GENERIF_SUMMARY	role of CDK4 in maintaining liposarcoma proliferation through its ability to inactivate RB function
Literature	GENERIF_SUMMARY	CDKN2A/RB1 increased in a stepwise manner from the normal mucosa, to esophagitis, dysplasia/carcinoma in situ, and carcinoma (P
Literature	GENERIF_SUMMARY	Binding of 53BP1 to methyl K810 occurs on E2 promoter binding factor target genes and allows pRb activity to be effectively integrated with the DNA damage response.
Literature	GENERIF_SUMMARY	The current study was conducted to query possible correlations among methylation statue of ATM gene, ATM/ RB protein expression, D1853N ATM polymorphism, telomere length, and clinicopathological characteristics of various types of brain tumors.
Literature	GENERIF_SUMMARY	no significant difference in expression between human papillomavirus-positive and -negative Bowen disease
Literature	GENERIF_SUMMARY	Rb94, which lacks the N-terminal 112 amino acid residues of the full-length protein, has a strong tumor-depressing effect in head and neck and bladder cancers.
Literature	GENERIF_SUMMARY	Invasive breast cancers are treated in distinct fashions, and heterogeneity within the RB pathway relates to prognosis and response to commonly used therapeutics.
Literature	GENERIF_SUMMARY	post-mitotic human cone precursors are uniquely sensitive to Rb depletion; Rb knockdown induced cone precursor proliferation in prospectively isolated populations and in intact retina
Literature	GENERIF_SUMMARY	Two RB1 mutations were found in the tumor; one RB1 mutation was present in 10% of blood cells, identifying mosaicsm.
Literature	GENERIF_SUMMARY	work identifies RABL6A as a novel negative regulator of

Main Category	Category	Terms
		Rb1
Literature	GENERIF_SUMMARY	lower levels of pRb and cyclin D1 and higher levels of E2F1, a member of the E2F family, were detected in tumors with RB1 alterations compared with tumors without RB1 alterations
Literature	GENERIF_SUMMARY	Suppression of genes within the RB and p53 pathways induces cell cycle reentry in intact human pancreatic islets.
Literature	GENERIF_SUMMARY	PRMT4 has a role in arginine methylation, which negatively regulates retinoblastoma tumor suppressor protein and promotes E2F-1 dissociation
Literature	GENERIF_SUMMARY	excess E2F1 due to Rb inactivation recruits the complex of Cdc20 and the anaphase-promoting complex/ cyclosome to deregulate the expression of UBCH10
Literature	GENERIF_SUMMARY	Our findings expand the spectrum of known RB1 mutations and will benefit the investigation of RB1 mutation hotspots. Genetic counseling can be offered to families with heritable RB1 mutations.
Literature	GENERIF_SUMMARY	High retinoblastoma protein expression is associated with chemosensitivity in hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	loss of RB1 protein expression by immunohistochemistry was seen in 4 four Merkel cell polyomavirus-negative primary salivary gland high-grade neuroendocrine carcinoma cases
Literature	GENERIF_SUMMARY	Data indicate that HPV16E7, CBP/p300, and retinoblastoma protein pRb interactions are potentially important for cellular transformation.
Literature	GENERIF_SUMMARY	RB maintains quiescence and prevents premature senescence through upregulation of DNMT1 in mesenchymal stromal cells
Literature	GENERIF_SUMMARY	These studies show that phosphorylation of Rb regulates its association with Bax and its role in apoptosis
Literature	GENERIF_SUMMARY	Cancer genome studies document recurrent genetic inactivation of RB1 and TP53, and the inactivation becomes more frequent in more advanced cancers. [Review]
Literature	GENERIF_SUMMARY	Epstein-Barr virus (EBV) infection is inversely correlated with pRb in Hodgkin/Reed-Sternberg (H/RS) cells in Classic Hodgkin lymphoma (cHL).
Literature	GENERIF_SUMMARY	co-transcription of ncRNA-RB1 and RB1 provides a positive link between the expression of the two tumor suppressors RB1 and the immune-relevant CALR protein
Literature	GENERIF_SUMMARY	There were seven individuals with germline mutations and all presented with advanced stage of tumor. There was no difference in age of onset of RB based on the germline mutation status.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	results indicate that Pin1 has a critical role in the modulation of Rb function by the regulation of Rb dephosphorylation, which may have an important pathological role in cancer development.
Literature	GENERIF_SUMMARY	Increased phosphorylated Rb is associated with Acute Lymphoblastic Leukemia.
Literature	GENERIF_SUMMARY	RB1 polymorphism contributes to the efficacy of platinum-taxanes in advanced squamous cell lung cancer.
Literature	GENERIF_SUMMARY	Authors found that retinoblastoma tumor suppressor knockdown prior to infection impaired human cytomegalovirus lytic infection at multiple stages, notably viral DNA replication, late protein expression, and infectious virion production.
Literature	GENERIF_SUMMARY	we re-tested a series of 40 patients with sporadic retinoblastoma
Literature	GENERIF_SUMMARY	RB is lost in resistant EGFR mutant lung adenocarcinomas that transform to small-cell lung cancer.
Literature	GENERIF_SUMMARY	The type and frequency of RB1 mutations in retinoblastoma patients were determined. Those who have deletions in the RB1 gene are at increased risk of chemoresistance and at higher risk of requiring enucleation.
Literature	GENERIF_SUMMARY	data identify RB1 loss as a candidate driver of structural genomic instability and a causative factor for cancer somatic heterogeneity and evolution
Literature	GENERIF_SUMMARY	We demonstrated for the first time that targeted next generation sequencing of rb1 protein is an efficient approach for the identification of wide spectrum of pathogenic variants in RB patients.
Literature	GENERIF_SUMMARY	For the subset of patients with heritable retinoblastoma-associated SNVs in the RB1 5' UTR, the absence of multiple structures is likely causative of the cancer
Literature	GENERIF_SUMMARY	RB-mediated glycolytic activation induces a metabolic flux into the oxidative phosphorylation pathway.
Literature	GENERIF_SUMMARY	Report association between human papillomavirus and Epstein - Barr virus DNA and gene promoter methylation of RB1 and CDH1 in range of cervical lesions.
Literature	GENERIF_SUMMARY	Coordinate pattern of Rb1+ and ER+ in HGSC and LGSC is 19 and 50%, respectively. Rb1 and P16 show inverse expression pattern according to tumor grade with more frequent Rb1 in low grade vs. more frequent P16 in grade 3 tumors
Literature	GENERIF_SUMMARY	Results show that specific Rb (S795) residue was significantly highly phosphorylated in esophageal adenocarcinoma associated with aberrant P16 expression in the cancer tissues.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Studies indicate that the non-hereditary form of retinoblastoma is caused by somatic mutations in the retinoblastoma RB1 gene, while the hereditary form is associated with germline mutations in the RB1 gene.
Literature	GENERIF_SUMMARY	An LXCXE motif in UL97 helps activate E2F-dependent promoters independently of its stimulatory effect on Rb phosphorylation.
Literature	GENERIF_SUMMARY	gene amplification, p53 and Rb disruption, and E2F1 activation drive FOXM1 expression in epithelial ovarian cancer, and FOXM1 promotes cell cycle progression in cell models
Literature	GENERIF_SUMMARY	RB1 mutation is associated with retinoblastoma.
Literature	GENERIF_SUMMARY	In this review, we discuss the role of the ubiquitin-proteasome system in proteolytic control of RB-E2F pathway components, and recent data that points to surprising non-proteolytic roles for the ubiquitin-proteasome system
Literature	GENERIF_SUMMARY	Expression rates of p16 and pRB differ according to the molecular subgroups of breast cancer and they subsequently correlate with clinicopathologic factors.
Literature	GENERIF_SUMMARY	Bradykinin inhibits oxidative stress-induced senescence of endothelial progenitor cells through the B2R/AKT/RB and B2R/EGFR/RB signal pathways.
Literature	GENERIF_SUMMARY	Deficiencies in Cellular Processes Modulated by the Retinoblastoma Protein Do Not Account for Reduced Human Cytomegalovirus Replication in Its Absence.
Literature	GENERIF_SUMMARY	Molecular screening and identification of the RB1 mutations in Pakistani patients discovered novel mutational variants of the RB1 gene in the Pakistani population.
Literature	GENERIF_SUMMARY	Loss of OTX2 expression resulted in decreased expression of C-MYC and CRX, genes previously implicated in retinoblastoma tumorigenesis. Loss of OTX2 expression increased the phosphorylation of RB, a potential mechanism of modulating cell proliferation
Literature	GENERIF_SUMMARY	Data show that alpha-linolenic acid (ALA) decreased the expression of VEGF and COX2 proteins, and reduced the expression of Human papillomavirus type 16/18 onco-proteins E6 and E7, resulting into restoration of expression of tumor suppressor proteins, p53 and Rb in cervical cancer cell lines.
Literature	GENERIF_SUMMARY	identified NRBE3 as a novel ubiquitin E3 ligase for RB that might play a role as a potential oncoprotein in human cancers
Literature	GENERIF_SUMMARY	p53 immunopositivity was more frequent in SCC (65%) than in VC (23%) (P
Literature	GENERIF_SUMMARY	Data indicate that single nucleotide polymorphisms (SNPs) in cyclin D1

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	By targeting genes involved in controlling pRb/E2F1 activity, miR-449a regulates cell cycle progression and apoptosis and consequently enhances the radiosensitivity of PC-3 cells. Thus, miR-449a, as a miRNA component of the Rb pathway, promotes the radiosensitivity of PC-3 cells through regulating pRb/E2F1.
Literature	GENERIF_SUMMARY	Spectrum of germ-line RB1 gene mutations has been described in Malaysian patients with retinoblastoma.
Literature	GENERIF_SUMMARY	UHRF1 promotes osteosarcoma cell invasion by downregulating the expression of Ecadherin and increasing epithelialtomesenchymal transition in an Rb1dependent manner.
Literature	GENERIF_SUMMARY	The activation of E2Fs is intimately regulated by retinoblastoma 1 (RB1). The RB pathway has been implicated in almost every human malignancy
Literature	GENERIF_SUMMARY	TGFbeta induces autophagy through the pRb/E2F1 pathway and transcriptional activation of autophagy-related genes
Literature	GENERIF_SUMMARY	These results indicate that the loss of RB1 and CDKN2A function induces distinctively different signaling cascades in non-small cell lung carcinoma (NSCLC) and small cell lung carcinoma
Literature	GENERIF_SUMMARY	Combinatorial PX-866 and Raloxifene Decrease Rb Phosphorylation, Cyclin E2 Transcription, and Proliferation of MCF-7 Breast Cancer Cells
Literature	GENERIF_SUMMARY	Rb knockdown reversed RBEL1 depletion-induced tumor suppressive effects. In conclusion, the present results suggest that RBEL1 modulates cell proliferation and G1S transition by inhibiting Rb in osteosarcoma
Literature	GENERIF_SUMMARY	Ras induces the formation of a complex between NORE1A and the phosphatase PP1A, promoting the activation of the Rb tumor suppressor by dephosphorylation
Literature	GENERIF_SUMMARY	This inhibition was accompanied by a strong decrease in G1 cyclins (particularly in cyclin D1) and retinoblastoma protein (Rb) phosphorylation
Literature	GENERIF_SUMMARY	The N terminus of the Retinoblastoma Protein (RB) inhibits DNA replication via interactions with Mcm7.
Literature	GENERIF_SUMMARY	These data collectively suggest a novel mechanism involving CSLC-specific targeting by evodiamine and its possible use to the therapy of breast cancer.
Literature	GENERIF_SUMMARY	These data suggest that CBX8 modulates SIPS through the RB-E2F1 pathway in CML cells and provide important insight into its application in CML treatment.
Literature	GENERIF_SUMMARY	efficacy of irinotecan on colorectal cancer could be predicted by phosphorylated RB expression, which is known to be crucial for G1/S progression and cell proliferation

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	We also observed that a dedifferentiated osteosarcomas (OS)harboring MDM2 amplification did not carry any other mutations. This study suggests that bone oncogenesis driven by TP53 or RB1 mutations occurs on a background of relative genetic stability and that the dedifferentiated OS subtype represents a clinico-pathological entity with distinct oncogenic mechanisms
Literature	GENERIF_SUMMARY	The two most frequently mutated genes were TP53 (86%) and RB1 (58%); other frequently mutated genes (>10% patients) were involved in epigenetic regulation as well as the mTOR pathway.We found that small-cell lung cancer tumors harboring no mutation in RB1 had a poor response to chemotherapy.
Literature	GENERIF_SUMMARY	findings support important role of trichodysplasia spinulosa-associated polyomavirus (TSPyV)-mediated pRb deactivation in development of trichodysplasia spinulosa (TS); observation that pRb tumor suppressor is inactivated by TSPyV sT but not Merkel cell polyomavirus sT provides further insights into the distinct pathobiological mechanisms of Merkel cell carcinoma and TS
Literature	GENERIF_SUMMARY	RB proteins stabilize interactions of the NuRD corepressor complex within the L1 promoter to effect L1 silencing
Literature	GENERIF_SUMMARY	Chk1's expression is controlled by p53 and RB/E2F1 at the transcriptional level.
Literature	GENERIF_SUMMARY	the previously reported CGI/retrocopy in intron 2 of the RB1 gene (PPP1R26P1) as well as the CGI associated with the retrocopy RPS2P32 identified in this study carry a maternal methylation imprint. In conclusion, these findings shed light on the evolutionary dynamics and constraints of DNA methylation.
Literature	GENERIF_SUMMARY	In pancreatic neoplasms, miR-137 targets KDM4A mRNA during Ras-induced senescence and activates both p53 and retinoblastoma (pRb) tumor suppressor pathways.
Literature	GENERIF_SUMMARY	We report the significance of genetic testing in the early detection and management of retinoblastoma from India.
Literature	GENERIF_SUMMARY	Specific E2Fs also have prognostic value in breast cancer, independent of clinical parameters. We discuss here recent advances in understanding of the RB-E2F pathway in breast cancer. We also discuss the application of genome-wide genetic screening efforts to gain insight into synthetic lethal interactions of CDK4/6 inhibitors in breast cancer for the development of more effective combination therapies.
Literature	GENERIF_SUMMARY	The maternally inherited c.1981C>T/p.Arg661Trp RB1 allele retains sufficient tumor suppressor activity to prevent retinoblastoma development.
Literature	GENERIF_SUMMARY	AR activity was inversely associated with cell

Main Category	Category	Terms
		proliferation, whereas the expression of Fa complex genes was correlated with elevated cell cycle progression, expression of the E2F transcription factor 1 (E2F1) and loss of RB1.
Literature	GENERIF_SUMMARY	the functional association of BLCAP and Rb1 might play important roles in proliferation and apoptosis of HeLa cells.
Literature	GENERIF_SUMMARY	these results demonstrate complex links between different signaling cascades in stressed hMESC, implicating ERK, p38 and JNK in regulation of the premature senescence via p53/p21/Rb pathway.
Literature	GENERIF_SUMMARY	RB underexpression is associated with tumor cell invasiveness and neuroendocrine differentiation in prostate cancer.
Literature	GENERIF_SUMMARY	Expression of Kv10.1 driven by phosphorylated Rb/E2F1 contributes to G2/M progression of cancer and non-transformed cells.
Literature	GENERIF_SUMMARY	miR-590 inhibits RB1 and promotes proliferation and invasion of T-cell acute lymphoblastic leukaemia cells
Literature	GENERIF_SUMMARY	Identify Dyrk1a as a novel negative regulator of D-cyclin-mediated Rb1/E2f-signalling. As dysregulation of this pathway with impaired cardiomyocyte proliferation leads to cardiomyopathy
Literature	GENERIF_SUMMARY	Authors show that MYC inhibition by Omomyc, a dominant-negative MYC, suppresses the growth of SCLC cells with TP53 and RB1 inactivation carrying MYC, MYCL, or MYCN amplification.
Literature	GENERIF_SUMMARY	Results suggest that RB1 is the dominant tumor suppressor PP in MCC, and that inactivation of RB1 by MCPyV-LT is largely sufficient for its growth supporting function in established MCPyV-positive MCC cells.
Literature	GENERIF_SUMMARY	the effect of low and medium dose radiation on mesenchymal stromal cells (MSCs) with inactivated RB1 gene, was investigated.
Literature	GENERIF_SUMMARY	AR mutations and/or copy number alterations were robustly detected in 48% (31 of 65) and 60% (18 of 30) of baseline and progression samples, respectively. Detection of AR amplification, heavily mutated AR (>/=2 mutations), and RB1 loss were associated with worse progression-free survival, with hazard ratios of 2.92
Literature	GENERIF_SUMMARY	a SUMO-deficient Rb mutant results in reduced SUMOylation and phosphorylation, weakened CDK2 binding, and attenuated E2F-1 sequestration.
Literature	GENERIF_SUMMARY	Inhibition of Rb phosphorylation by depleting cyclin D or using CDK4/6 inhibitors releases Rb-mediated mTORC2 suppression. This, in turn, leads to elevated Akt activation to confer resistance to chemotherapeutic drugs in Rb-proficient cells, which can be attenuated with Akt inhibitors.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Suppression of tumorigenesis by CADM1 may be mediated by the Rb-E2F1 pathway.
Literature	GENERIF_SUMMARY	Data suggest that the platelet derived growth factor receptor alpha (PDGFRalpha)/Stat3 transcription factor/Rb1 protein regulatory axis might represent a potential therapeutic target for glioblastoma (GBM) treatment.
Literature	GENERIF_SUMMARY	The expression of Rb and MeCP2 in patients with B-cell non-Hodgkin's lymphoma (B-NHL) showed that positive staining for MeCP2 or Rb was significantly lower in B-NHL tumor tissues, and these changes were significantly and negatively correlated with the grade of B-NHL.
Literature	GENERIF_SUMMARY	APC/C and pRB interact with each other via the co-activator of APC/C, FZR1, providing an alternative pathway of regulation of G1 to S transition by pRB using a post-translational mechanism. Both pRB and FZR1 have complex roles and are implicated not only in regulation of cell proliferation but also in differentiation, quiescence, apoptosis, maintenance of chromosomal integrity and metabolism.
Literature	GENERIF_SUMMARY	CTDSPL and Rb directly interact and can be involved in the common mechanism of cell cycle regulation.
Literature	GENERIF_SUMMARY	The authors conclude that human cytomegalovirus can obtain deoxyribonucleotide triphosphates in the absence of Rb phosphorylation and that viral UL97 can contribute to the efficiency of DNA replication in an Rb phosphorylation-independent manner.
Literature	GENERIF_SUMMARY	TYROBP influences a batch of genes that are related to Alzheimer's disease; ZNF329 and RB1 significantly regulate those 'mesenchymal' gene expression signature genes for brain tumors. By merely leveraging gene expression data, Context Based Dependency Network (CBDN) can efficiently infer the existence of gene-gene interactions as well as their regulatory directions.
Literature	GENERIF_SUMMARY	RB1 deficiency promotes cancer cell proliferation in part by enhancing mitochondrial function and identify tigecycline as a clinically approved drug for RB1-deficient triple negative breast cancer.
Literature	GENERIF_SUMMARY	Results suggest that retinoblastoma 1 (RB1) promoter methylation may be associated with GC and H. pylori infection.
Literature	GENERIF_SUMMARY	this paper shows that patients with estrogen receptor positive breast cancer expressing high Rb loss-of-function have significantly worse recurrence free survival compared to those with low Rb loss-of-function
Literature	GENERIF_SUMMARY	The N-Terminal phosphorylation of RB by p38 bypasses its inactivation by cyclin-dependent kinases and prevents proliferation in cancer cells.
Literature	GENERIF_SUMMARY	Data suggest that gamma-tocotrienol delays cellular

Main Category	Category	Terms
		senescence of cultured diploid fibroblasts by regulating the cell cycle through down-regulation of expression of p16-INK4a (CDKN2A/cyclin dependent kinase inhibitor 2A) mRNA, of expression/phosphorylation of retinoblastoma protein (RB1), and of expression of cyclin D1 (CCND1).
Literature	GENERIF_SUMMARY	AR also indirectly increases the expression of DNA replication genes through stimulatory effects on other metabolic genes with subsequent CDK activation and Rb hyperphosphorylation.
Literature	GENERIF_SUMMARY	Here, we introduce a transcriptional signature to specifically track CDK2 activity. It responds to genetic and chemical perturbations in the CDK-RB-E2F axis, correlates with mitotic rate in vitro and in vivo and reacts rapidly to changes in CDK2 activity during cell cycle progression
Literature	GENERIF_SUMMARY	Suggest frequent Rb1 deficiency as a possible driver molecular event in superficial acral fibromyxoma.
Literature	GENERIF_SUMMARY	we investigated how Rb is integrated to control bladder cancer progression through E2F3 and p53 regulation. The results exhibit that Rb expression is lower in patients with bladder tumor, while E2F3 level is high. Rb knockdown enhanced bladder tumor cell proliferation and migration, aggravated with p53 silence.
Literature	GENERIF_SUMMARY	RB loss drives metastatic capacity and highlight RHAMM as a candidate therapeutic target for treating advanced prostate cancer.
Literature	GENERIF_SUMMARY	TP53 and RB1 mutations seem to be restricted to high-risk/malignant gastrointestinal stromal tumors and occur at an equal although relatively low frequency.
Literature	GENERIF_SUMMARY	Increased the expression of tumor suppressor genes such as Cdkn2A, Rb1 and p53.
Literature	GENERIF_SUMMARY	We further discuss how dysfunction in the p53- and/or Rb-mediated mechanosignaling pathways is potentially involved in cancer progression. These pathways might provide good targets for developing anticancer therapies
Literature	GENERIF_SUMMARY	these results identify crucial functions for RB in the cerebral organoid model of human brain development.
Literature	GENERIF_SUMMARY	Statistically significant correlations were found only in Morbus Bowen (preinvasive cutaneous carcinoma) between parameters HPV-p53, p53-pRb and p53-p16.
Literature	GENERIF_SUMMARY	The SNPs rs 216311, rs 1800383 and rs 1800386 associated significantly with bleeding in study subjects. rs1800386 occurred in all with bleeding history, no ethnic variations were noted.
Literature	GENERIF_SUMMARY	This review focuses on the ability of pRB to control nucleosome/chromatin structures via physical interactions with histone modifiers and chromatin factors and describes cancer therapies based on

Main Category	Category	Terms
		targeting these protein factors.
Literature	GENERIF_SUMMARY	SOX2 overexpression and the loss of Rb1 protein expression might have a pivotal role in the divergent differentiation of pluripotent embryonic-like epithelial cells and the development of esophageal small-cell carcinoma.
Literature	GENERIF_SUMMARY	The Rb1 tumor suppressor gene modifies telomeric chromatin architecture by regulating TERRA expression.
Literature	GENERIF_SUMMARY	the frequency and association of polymorphisms in the TP53 and RB1 genes with clinical characteristics in a group of children with retinoblastoma (RB) in northern Mexico, was examined.
Literature	GENERIF_SUMMARY	Results show that the functional state of protein Rb is inferred to be inactive due its phosphorylation status in the MYCN-amplified retinoblastoma without coding sequence mutations. This makes inactivation of RB1 by gene mutation or by protein phosphorylation, a necessary condition for initiating retinoblastoma tumorigenesis, independent of MYCN amplification.
Literature	GENERIF_SUMMARY	homozygous loss of RB1 is an independent prognostic marker in multiple myeloma
Literature	GENERIF_SUMMARY	A relatively stable genome in retinoblastoma tumor cells is maintained by TRb1 and TRb2-mediated PTTG1 inhibition, counteracting Rb-deficiency-related genomic instability.
Literature	GENERIF_SUMMARY	The epigenetic interaction between Linc00441 and bidirectional transcripted neighbor RB1 may be a de novo theory cutting-point for the inactivation of RB1 in HCC.
Literature	GENERIF_SUMMARY	Rb gene promoter methylation was more frequent in gastric cancer patients than in controls.
Literature	GENERIF_SUMMARY	Studies suggest that mitochondrial ribosomal protein S18-2 protein, together with the retinoblastoma protein (RB), plays a crucial role in the control on cell stemness and differentiation
Literature	GENERIF_SUMMARY	Analyses with phospho-defective and phospho-mimetic mutants of FoxM1b identified a critical role of the Plk1 phosphorylation sites in regulating the binding of FoxM1b to Rb and DNMT3b.
Literature	GENERIF_SUMMARY	several RB1 alterations associated to retinoblastoma in the human were present in several non-human primates without an apparent pathological effect.
Literature	GENERIF_SUMMARY	to our knowledge, the first report of miR-26a targeting Rb1 in colon cancer. The results of this study suggested that miR-26a could serve as a progression biomarker in colorectal cancer. Further validation studies are still needed to confirm our findings.
Literature	GENERIF_SUMMARY	Loss of Rb immunolabeling and KRAS mutation are promising molecular markers of the therapeutic

Main Category	Category	Terms
		response to platinum-based chemotherapy for pancreatic neuroendocrine neoplasm grade-3 (PanNEN-G3), and Rb for neuroendocrine tumor with G3 (NET-G3).
Literature	GENERIF_SUMMARY	Low RB expression is associated with breast cancer.
Literature	GENERIF_SUMMARY	Aberrant expressionof Rb is frequently occurred in lung adenocarcinoma patients with EGFR mutations.
Literature	GENERIF_SUMMARY	Loss of retinoblastoma in pleomorphic fibroma
Literature	GENERIF_SUMMARY	The p53-mediated sensitivity of breast cancer cells to chemotherapeutic agents appears to be driven mostly by pRB. Using agents that enhance RB phosphorylation might possibly increase the chemosensitivity of breast cancer cells.
Literature	GENERIF_SUMMARY	RB1 was identified as a direct and functional target of miR-215. RB1 is generally down-regulated in glioma tissues and its expression inversely correlated with miR-215, which is up-regulated in high-grade glioma tissues, and its expression was negatively correlated with miR-215.
Literature	GENERIF_SUMMARY	This is the first published report on the spectrum of mutations in RB patients from Singapore and shows that further improved mutation screening strategies are required in order to provide a definitive molecular diagnosis for every case of RB
Literature	GENERIF_SUMMARY	results define a network of E2F target genes as susceptible to the regulatory influence of H1.2, where H1.2 augments global association of pRb with chromatin, enhances transcriptional repression by pRb, and facilitates pRb-dependent cell-cycle arrest
Literature	GENERIF_SUMMARY	Low RB expression is associated with osteosarcoma.
Literature	GENERIF_SUMMARY	We recommend intensive ocular screening for patients with germline RB1 mutations for retinoblastoma as well as neuroimaging for pineoblastoma surveillance. There is an approximately 20% risk of developing second primary cancers among individuals with hereditary RB, higher among those who received radiotherapy for their primary RB tumors
Literature	GENERIF_SUMMARY	miR-215 promoted cell migration and invasion of gastric cancer by directly targeting RB1.
Literature	GENERIF_SUMMARY	MiR-661 promotes metastasis of non small cell lung cancer through RB/E2F1 signaling and epithelial-mesenchymal transition events.
Literature	GENERIF_SUMMARY	Loss of RB1 is associated with papillomavirus involvement in Barrett's dysplasia and esophageal adenocarcinoma.
Literature	GENERIF_SUMMARY	Analysis of the spectrum of RB1 variants observed in 60,706 exomes identifies 197 variants that have enough potential to disrupt splicing to warrant further consideration.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Disruption of DREAM and RB-E2F complexes by oncoproteins from DNA tumor viruses leads to upregulation of cell cycle genes and impairs growth-inhibiting pathways, including the p53-mediated downregulation of cell cycle genes. [review]
Literature	GENERIF_SUMMARY	Data indicate that nuclear envelope rupture in cancer cells is likely due to loss of either the Rb or the p53 pathway.
Literature	GENERIF_SUMMARY	RB inactivation enhances pro-inflammatory signaling through stimulation of the interleukin-6/STAT3 pathway, which directly promotes various malignant features of cancer cells. [review]
Literature	GENERIF_SUMMARY	The authors report that a subset of cell cycle-related genes including retinoblastoma 1 is the target of Rbfox2 in cytoplasmic stress granules, and Rbfox2 regulates the retinoblastoma 1 mRNA and protein expression levels during and following stress exposure.
Literature	GENERIF_SUMMARY	Altered pRb is frequently expressed in gastric carcinoma, inversely correlates with tumor invasion and tumor stage suggesting an early event in gastric carcinogenesis.
Literature	GENERIF_SUMMARY	Expression levels of miR-675-5p in glioma tissues and cells were negatively correlated with RB1 expression at both mRNA and protein levels and promoted cell proliferation and migration.
Literature	GENERIF_SUMMARY	In certain contexts, Rb loss enables TRbeta1-dependent suppression of SKP2 as a safeguard against RB1-deficient tumorigenesis. TRbeta2 counteracts TRbeta1, thus disrupting this safeguard and promoting development of RB1-deficient malignancies.
Literature	GENERIF_SUMMARY	The data indicate that MAZ is essential to bypass MYB promoter repression by RB family members and to induce MYB expression.
Literature	GENERIF_SUMMARY	Concurrent HR deficiency and RB1 loss were associated with favorable outcomes, suggesting that co-occurrence of specific mutations might mediate durable responses in such patients
Literature	GENERIF_SUMMARY	In conclusion, a novel low penetrance deep intronic variant within RB1 intron 18 was discovered (c.1814 + 1307C>G). Confirmation of the RB1 pathogenic variant and the low penetrance status assisted in surveillance scheduling and pregnancy management for this family.
Literature	GENERIF_SUMMARY	Reduced RB expression in medullary thyroid cancer is associated with decreased patient survival in univariate and multivariable analyses, independent from patient age at surgery or advanced TNM stage.
Literature	GENERIF_SUMMARY	he present result indicated that vascular smooth proliferation is regulated by activation of the NF-kappaB p65/miR17/RB pathway. As NF-kappaB p65 signalling is activated in and is a master regulator of the inflammatory response, the present findings may

Main Category	Category	Terms
		provide a mechanism for the excessive proliferation of VSMCs under inflammation during vascular disorders and may identify novel targets for the treatment of vascular d...
Literature	GENERIF_SUMMARY	According to immunohistochemistry and immunoblot analysis, the expression levels of cyclin D1, cyclin E, pRb, and Ki67 in psoriasis lesions decreased after treatment and were similar with those in the normal group
Literature	GENERIF_SUMMARY	The increased expression of miR-503-5p significantly reduced the expressions of E2F transcription factor 3 (E2F3) mRNA and retinoblastoma protein (Rb)/E2F signaling pathway mRNA in bladder cancer cells.
Literature	GENERIF_SUMMARY	these findings not only identify tumor-suppressive functions of RB that are distinct from cell cycle control, but also demonstrate that the molecular consequence of RB loss is distinct from RB inactivation.
Literature	GENERIF_SUMMARY	causative RB1 mutations in most bilateral retinoblastoma (RB) patients and in some unilateral RB patients, including five novel mutations, were identified.
Literature	GENERIF_SUMMARY	miR-181a plays an oncogenic role in thyroid cancer; by targeting RB1, it promotes cell cycle progression and inhibits apoptosis.
Literature	GENERIF_SUMMARY	Mutation in RB1 gene is not associated with age at diagnosis for unilateral retinoblastoma.
Literature	GENERIF_SUMMARY	Results show that RB1 expression is regulated by cdc37 which facilitates its phosphorylation through increasing CDK4 stability.
Literature	GENERIF_SUMMARY	BMP-2 is a driving factor for promoting epithelial mesenchymal transition and breast cancer stemness via Rb and CD44 signaling pathways
Literature	GENERIF_SUMMARY	Concurrent mutations, in genes such as CDKN2B or RB1, were associated with worse clinical outcome in lung adenocarcinoma patients with EGFR active mutations.
Literature	GENERIF_SUMMARY	The two most common deletions were tumor-suppressor genes RB1 and ERBB4 in chromophobe renal cell carcinoma but not renal oncocytoma.
Literature	GENERIF_SUMMARY	this study shows that pRb loss emerged as the most significant poor prognostic indicator in conjunctival squamous cell carcinoma
Literature	GENERIF_SUMMARY	Mutational screening of germline RB1 gene in Vietnamese patients with retinoblastoma reveals three novel mutations.
Literature	GENERIF_SUMMARY	PPM1B plays a negative role in the activation of the p38-RB1-E2F1 pathway and that targeting PPM1B could be useful in certain types of cancer by stimulating chemotherapy-induced cell death.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	3-fold excess of RB1 maternal canonical transcript led us to hypothesize that this differential allelic expression could influence phenotypic outcome in families at risk for Retinoblastoma onset
Literature	GENERIF_SUMMARY	E3 ubiquitin ligase RNF123 targets LMNB1, Rb protein and LAP2alpha for proteasomal degradation.
Literature	GENERIF_SUMMARY	Expression of aspartate beta-hydroxylase (ASPH) promoted direct protein interaction between RB1, cyclin-dependent kinases, and cyclins.
Literature	GENERIF_SUMMARY	Promoter hypermethylation was detected in 11.4% of the retinoblastoma tumors and should be tested for in routine RB1 screening programmes.
Literature	GENERIF_SUMMARY	RASSF6 increases unphosphorylated pRb and augments the interaction between pRb and E2F1. Moreover, RASSF6 induces TP73 target genes via pRb and E2F1 in a p53-negative background.
Literature	GENERIF_SUMMARY	Overexpression of TRAF3 in HPV(+) cell lines with decreased endogenous TRAF3 inhibited NF-kappaB2/RELB expression, nuclear localization, and NF-kappaB reporter activity, while increasing the expression of IFNA1 mRNA and protein and sensitizing cells to its growth inhibition. Overexpression of TRAF3 also enhanced TP53 and RB tumor suppressor proteins and decreased HPV E6 oncoprotein in HPV(+) cells
Literature	GENERIF_SUMMARY	results showed that a) alterations of the p53 and Rb pathways are associated with high proliferation of tumor cells in BUC and b) high expression of cell-cycle proteins is associated with adverse histopathological parameters of these tumors
Literature	GENERIF_SUMMARY	No correlation was found between RB1germline mutations and second primary malignancies in hereditary retinoblastoma. EBRT treated patients with a low penetrance RB1 mutation remain at risk of second primary malignancies.
Literature	GENERIF_SUMMARY	vitiligo lesions exhibited dysregulated SUMOylation and deSUMOylation in keratinocytes, dysregulation of the cell cycle progression was observed in SUMO1 knockdown HaCaT cells and the deSUMOylation of Rb in keratinocytes may serve an important role in the development of vitiligo.
Literature	GENERIF_SUMMARY	Control of the Restriction Point by Rb and p21.
Literature	GENERIF_SUMMARY	These findings demonstrate that developmental stage-specific as well as species- and cell type-specific features sensitize to RB1 inactivation and reveal the human cone precursors' capacity to model retinoblastoma initiation, proliferation, premalignant arrest, and tumor growth.
Literature	GENERIF_SUMMARY	mCRPCs have a high prevalence of RB1 genomic aberrations, with structural variants, including rearrangements, being common. Intrapatient genomic and expression heterogeneity favors RB1 aberrations as

Main Category	Category	Terms
		late, subclonal events that increase in prevalence due to treatment-selective pressures.
Literature	GENERIF_SUMMARY	Low pRB expression is associated with mouth Cancer.
Literature	GENERIF_SUMMARY	The accumulation of sequence variations in RB1 gene might influence Greek patients' susceptibility towards the progression of cervical neoplasia.
Literature	GENERIF_SUMMARY	miR-199a-3p is down-regulated in rheumatoid arthritis (RA)-Fibroblast-like synoviocytes (FLSs)(RA-FLS) and miR-199a-3p inhibits proliferation and induces apoptosis in , partially via targetting RB1. The miR-199a-3p/RB1 pathway may represent a new therapeutic target for RA.
Literature	GENERIF_SUMMARY	RB1 Loss Is Synthetic Lethal with Aurora B Kinase That Regulate Chromosomal Segregation in Small cell lung cancer.
Literature	GENERIF_SUMMARY	We delineate two potentially distinct molecular-genetic subtypes of OA sebaceous carcinoma. The first is defined by somatic mutations impacting TP53 and/or RB1 [20/29 (70%) patients, including 10 patients whose primary tumors contained coexisting TP53 and RB1 mutations] with frequent concomitant mutations affecting NOTCH genes.
Literature	GENERIF_SUMMARY	Using a tag-SNP, methylation was demonstrated to affect the maternal allele. Real-time qPCR demonstrated RB1 transcriptional silencing. In conclusion, we documented that promoter methylation can act as the first ""hit"" in Knudson's model"
Literature	GENERIF_SUMMARY	a biomarker consisting of the phosphorylation of the retinoblastoma protein (Rb) on serine 249 combined with elevated p39 expression. This biomarker correlates with epithelial-to-mesenchymal transition traits in non-small cell lung carcinoma (NSCLC) cells.
Literature	GENERIF_SUMMARY	the effects of combination treatment of phosphatase activation with TKIs on cell number and activation of the signal transducer and activator of transcription 3 (STAT3) resistance pathway were determined. The combination of Rb phosphatase activation with TKIs resulted in a greater reduction in cell number compared with either treatment alone, without STAT3 pathway activation.
Literature	GENERIF_SUMMARY	Deleterious variants are associated with an increased risk of relapse and death in Castration-Sensitive Prostate Cancer . Poorer outcomes are seen with compound gene hits in both early and advanced disease, and this may in part reflect increasing global genomic instability.
Literature	GENERIF_SUMMARY	Spectrum of germline RB1 mutations and clinical manifestations in retinoblastoma patients from Thailand have been reported.
Literature	GENERIF_SUMMARY	RB activation signals are integrated in a phosphorylation code that determines the diversity of

Main Category	Category	Terms
		RB activity.
Literature	GENERIF_SUMMARY	these findings reveal an important and coordinated role for RB1 and CDKN2a in regulating cell-cycle progression and differentiation during human cardiomyogenesis. transient, dual inactivation of RB1 and CDKN2a in endogenous adult GATA4(+) cells and cardiomyocytes mediates, at least in part, the beneficial effects of cell-based therapy in a post-MI large mammalian model, a finding with potential clinical implications.
Literature	GENERIF_SUMMARY	Results found RB1 mutant status in 8.2% of a cohort of patients with non small cell lung cancer and strongly associated with worse outcomes.
Literature	GENERIF_SUMMARY	The expression of E6 and L1 is markedly lower, and expression of p53 and Rb is significantly higher after REE treatment in HFK-HPV18 or organotypic rafts containing HFK-HPV18. Among all formula of REE, ethanol-extracted compound REE displays the highest protection against HPV18
Literature	GENERIF_SUMMARY	Our work conclusively demonstrates that the cyclin D-Rb interaction drives cell division and expands the diversity of known cyclin-based protein docking mechanisms.
Literature	GENERIF_SUMMARY	Mutations in the RB1 gene in Vietnamese patients with retinoblastoma were heterogeneous and highly prevalent with pathogenic truncated mutations.
Literature	GENERIF_SUMMARY	findings suggest that altered p53, p16, and/or Rb expression is relatively specific to porocarcinoma in comparison with poroma.
Literature	GENERIF_SUMMARY	RB constrains lineage fidelity and multiple stages of tumour progression and metastasis
Literature	GENERIF_SUMMARY	Genomic profiling of ovarian cancer patients revealed that response rate to platinum retreatment was 22% in patients with CCNE1 gains and 83.5% in patients with no CCNE1 gains. Response rate was 54.5% in patients with RB1 loss and 25% in patients without RB1 loss. CCNE1 gain and RB1 loss discriminate patients with tumors extremely sensitive to platinum retreatment.
Literature	GENERIF_SUMMARY	Results present evidence for increased DNA damage and instability in cancer cells with RB pathway defects when RB1 mutations are induced. Also, late-arising RB1 mutations can facilitate genome instability and cancer progression that are beyond the preexisting proliferative control deficit.
Literature	GENERIF_SUMMARY	we proposed a model for a SOX4/miR-17-92/RB1 axis which gives insight into how RB1 protein is reduced (besides RB1 loss). Mechanically, SOX4 downregulates RB1 protein expression by upregulating the miR-17-92 cluster expression.
Literature	GENERIF_SUMMARY	The mitochondrial Drp1-RB axis would be a novel target for the protection cells from xenobiotics triggering

Main Category	Category	Terms
		hepatic injury and diseases involved in necroptosis.
Literature	GENERIF_SUMMARY	Inhibition of Senescence-Associated Genes Rb1 and Meis2 in Adult Cardiomyocytes Results in Cell Cycle Reentry and Cardiac Repair Post-Myocardial Infarction.
Literature	GENERIF_SUMMARY	p53-dependent repression of cell-cycle genes is completely abrogated in LIN37-/-;RB-/- cells leading to a loss of the G1/S checkpoint.
Literature	GENERIF_SUMMARY	our findings imply allele-specific transcriptional regulation of HDAC9 via E2F3 and Rb1 as a major mechanism mediating vascular risk at rs2107595.
Literature	GENERIF_SUMMARY	Rb-E2F bistable switch controls cell cycle entry. (Review)
Literature	GENERIF_SUMMARY	Genetic testing identified a low-penetrance germline mutation in RB1 [NM_000321.2
Literature	GENERIF_SUMMARY	Phosphoproteomic and Kinomic Signature of Clinically Aggressive Grade I (1.5) Meningiomas Reveals RB1 Signaling as a Novel Mediator and Biomarker.
Literature	GENERIF_SUMMARY	The HPV cases of sinonasal squamous cell carcinoma (SNSCC) were located predominantly at the nasal cavity with nonkeratinizing histology and partial loss of Rb. Notably, 40% (6/15) of p16 SNSCCs were HPV. Two of these cases showed complete loss of Rb expression by immunohistochemistry, suggesting a reason for the above discrepancy.
Literature	GENERIF_SUMMARY	REVIEW
Literature	GENERIF_SUMMARY	Genome-wide profiling of non-smoking-related lung cancer cells reveals common RB1 rearrangements associated with histopathologic transformation in EGFR-mutant tumors.
Literature	GENERIF_SUMMARY	Unilateral retinocytoma associated with a variant in the RB1 gene.
Literature	GENERIF_SUMMARY	Silencing of KIF3A suppressed TNBC cell proliferation by repressing the Rb-E2F signaling pathway and inhibited migration and invasion by repressing epithelial-mesenchymal transition. The tumor size was smaller and the number of lung metastatic nodules was lower in KIF3A depletion MDA-MB-231 cell xenograft mice than in the negative control group. In addition, KIF3A overexpression correlated with chemoresistance.
Literature	GENERIF_SUMMARY	MicroRNA-192 promotes the development of nasopharyngeal carcinoma through targeting RB1 and activating PI3K/AKT pathway.
Literature	GENERIF_SUMMARY	Urothelial Cancers with Small Cell Variant Histology Have Confirmed High Tumor Mutational Burden, Frequent TP53 and RB Mutations, and a Unique Gene Expression Profile.
Literature	GENERIF_SUMMARY	BRCAness, SLFN11, and RB1 loss predict response to topoisomerase I inhibitors in triple-negative breast

Main Category	Category	Terms
		cancers.
Literature	GENERIF_SUMMARY	Deregulation of cell growth and apoptosis in UV-induced melanomagenesis.
Literature	GENERIF_SUMMARY	NRF3-POMP-20S Proteasome Assembly Axis Promotes Cancer Development via Ubiquitin-Independent Proteolysis of p53 and Retinoblastoma Protein.
Literature	GENERIF_SUMMARY	Next-Generation Sequencing of Retinoblastoma Identifies Pathogenic Alterations beyond RB1 Inactivation That Correlate with Aggressive Histopathologic Features.
Literature	GENERIF_SUMMARY	Germline RET variants underlie a subset of paediatric osteosarcoma.
Literature	GENERIF_SUMMARY	Role of RB1 in human embryonic stem cell-derived retinal organoids.
Literature	GENERIF_SUMMARY	Genomic profiling reveals the pivotal role of hrHPV driving copy number and gene expression alterations, including mRNA downregulation of TP53 and RB1 in penile cancer.
Literature	GENERIF_SUMMARY	Impact of RB1 gene mutation type in retinoblastoma patients on clinical presentation and management outcome.
Literature	GENERIF_SUMMARY	Tumor Milieu Controlled by RB Tumor Suppressor.
Literature	GENERIF_SUMMARY	Study reports that the proteasome cannot initiate degradation efficiently on full-length Rb. Calpain cleavage exposes a region that is recognized by the proteasome, leading to rapid proteolysis of Rb. These findings identify a mechanism for regulating protein stability by controlling initiation and provide a better understanding of the molecular mechanism underlying transformation by human papilloma virus.
Literature	GENERIF_SUMMARY	Genetic and expression variations of cell cycle pathway genes in brain tumor patients.
Literature	GENERIF_SUMMARY	The impact of RB1 genotype on incidence of second tumours in heritable retinoblastoma.
Literature	GENERIF_SUMMARY	Integrated case-control and somatic-germline interaction analyses of soft-tissue sarcoma.
Literature	GENERIF_SUMMARY	Regulating tumor suppressor genes
Literature	GENERIF_SUMMARY	CRISPR/Cas9 treatment causes extended TP53-dependent cell cycle arrest in human cells.
Literature	GENERIF_SUMMARY	Pharmacologically targetable vulnerability in prostate cancer carrying RB1-SUCLA2 deletion.
Literature	GENERIF_SUMMARY	Chromosomal microarray analysis of benign mesenchymal tumors with RB1 deletion.
Literature	GENERIF_SUMMARY	GLUT1 inhibition blocks growth of RB1-positive triple negative breast cancer.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Prognostic Information for Known Genetic Carriers of RB1 Pathogenic Variants (Germline and Mosaic).
Literature	GENERIF_SUMMARY	MicroRNA-215 promoted the progression of nasopharyngeal carcinoma through targeting RB1 and activating Wnt/beta-catenin pathway.
Literature	GENERIF_SUMMARY	Heterozygous retinoblastoma gene mutation compromises in vitro osteogenesis of adipose mesenchymal stem cells - a temporal gene expression study.
Literature	GENERIF_SUMMARY	NIR promotes progression of colorectal cancer through regulating RB.
Literature	GENERIF_SUMMARY	FGFR1 Is Critical for RBL2 Loss-Driven Tumor Development and Requires PLCG1 Activation for Continued Growth of Small Cell Lung Cancer.
Literature	GENERIF_SUMMARY	Synthetic lethality of RB1 and aurora A is driven by stathmin-mediated disruption of microtubule dynamics.
Literature	GENERIF_SUMMARY	Chimerism involving a RB1 pathogenic variant in monochorionic dizygotic twins with twin-twin transfusion syndrome.
Literature	GENERIF_SUMMARY	Next generation sequencing of RB1gene for the molecular diagnosis of ethnic minority with retinoblastoma in Yunnan.
Literature	GENERIF_SUMMARY	DNA hypermethylation/boundary control loss identified in retinoblastomas associated with genetic and epigenetic inactivation of the RB1 gene promoter.
Literature	GENERIF_SUMMARY	The Role of Retinoblastoma Protein in Cell Cycle Regulation
Literature	GENERIF_SUMMARY	HDAC5 Loss Impairs RB Repression of Pro-Oncogenic Genes and Confers CDK4/6 Inhibitor Resistance in Cancer.
Literature	GENERIF_SUMMARY	Spectrum of germline mutations in RB1 in Chinese patients with retinoblastoma
Literature	GENERIF_SUMMARY	Clinical characteristics and germline mutation spectrum of RB1 in Chinese patients with retinoblastoma
Literature	GENERIF_SUMMARY	Whole-exome mutational landscape of neuroendocrine carcinomas of the gallbladder.
Literature	GENERIF_SUMMARY	Cross-species identification of PIP5K1-, splicing- and ubiquitin-related pathways as potential targets for RB1-deficient cells.
Literature	GENERIF_SUMMARY	FOXP1 drives osteosarcoma development by repressing P21 and RB transcription downstream of P53.
Literature	GENERIF_SUMMARY	Comprehensive analysis of mutational and clinicopathologic characteristics of poorly differentiated colorectal neuroendocrine carcinomas.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	Roles of the microRNA3383p/NOVA1 axis in retinoblastoma.
Literature	GENERIF_SUMMARY	Relationships between highly recurrent tumor suppressor alterations in 489 leiomyosarcomas.
Literature	GENERIF_SUMMARY	Secondary enucleated retinoblastoma with MYCN amplification.
Literature	GENERIF_SUMMARY	Glioblastomas with primitive neuronal component harbor a distinct methylation and copy-number profile with inactivation of TP53, PTEN, and RB1.
Literature	GENERIF_SUMMARY	RB1 and TP53 co-mutations correlate strongly with genomic biomarkers of response to immunity checkpoint inhibitors in urothelial bladder cancer.
Literature	GENERIF_SUMMARY	MiR-3663-3p Inhibits the Progression of Gastric Cancer Through the CCND1 Pathway.
Literature	GENERIF_SUMMARY	The nucleoplasmic interactions among Lamin A/C-pRB-LAP2alpha-E2F1 are modulated by dexamethasone.
Literature	GENERIF_SUMMARY	RABL6A Regulates Schwann Cell Senescence in an RB1-Dependent Manner.
Literature	GENERIF_SUMMARY	Correlation between expression levels of lncRNA FER1L4 and RB1 in patients with colorectal cancer.
Literature	GENERIF_SUMMARY	Multi-omics profiling of primary small cell carcinoma of the esophagus reveals RB1 disruption and additional molecular subtypes.
Literature	GENERIF_SUMMARY	RB1 germline mutation spectrum and clinical features in patients with unilateral retinoblastomas.
Literature	GENERIF_SUMMARY	Differential impact of tumor suppressor gene (TP53, PTEN, RB1) alterations and treatment outcomes in metastatic, hormone-sensitive prostate cancer.
Literature	GENERIF_SUMMARY	MiR-449a Sponges Cyclin D1 Expression and Suppresses Phosphorylation of Retinoblastoma Protein against Osteosarcoma.
Literature	GENERIF_SUMMARY	NGS Analysis Confirms Common TP53 and RB1 Mutations, and Suggests MYC Amplification in Ocular Adnexal Sebaceous Carcinomas.
Literature	GENERIF_SUMMARY	Multi-region exome sequencing reveals the intratumoral heterogeneity of surgically resected small cell lung cancer.
Literature	GENERIF_SUMMARY	Cancer cells with defective RB and CDKN2A are resistant to the apoptotic effects of rapamycin.
Literature	GENERIF_SUMMARY	Identification of novel RB1 genetic variants in Retinoblastoma patients and their impact on clinical outcome.
Literature	GENERIF_SUMMARY	The Retinoblastoma Tumor Suppressor Is Required for the NUP98-HOXA9-Induced Aberrant Nuclear Envelope Phenotype.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	SUMOylation regulates Rb hyperphosphorylation and inactivation in uveal melanoma.
Literature	GENERIF_SUMMARY	Alterations in the RB Pathway With Inactivation of RB1 Characterize Glioblastomas With a Primitive Neuronal Component.
Literature	GENERIF_SUMMARY	Multiomics Analysis of Spatially Distinct Stromal Cells Reveals Tumor-Induced O-Glycosylation of the CDK4-pRB Axis in Fibroblasts at the Invasive Tumor Edge.
Literature	GENERIF_SUMMARY	miR-222-5p promotes dysfunction of human vascular smooth muscle cells by targeting RB1.
Literature	GENERIF_SUMMARY	Retinoblastoma protein regulates carcinogen susceptibility at heterochromatic cancer driver loci.
Literature	GENERIF_SUMMARY	Analysis of the Frequency of 10 Polymorphic Markers of CDKN2A and RB1 Genes in Russian Populations.
Literature	GENERIF_SUMMARY	Active RB causes visible changes in nuclear organization.
Literature	GENERIF_SUMMARY	Functional In Vitro Assessment of VEGFA/NOTCH2 Signaling Pathway and pRB Proteasomal Degradation and the Clinical Relevance of Mucolipin TRPML2 Overexpression in Glioblastoma Patients.
Literature	GENERIF_SUMMARY	Mutually exclusive mutation profiles define functionally related genes in muscle invasive bladder cancer.
Literature	GENERIF_SUMMARY	RB1 loss overrides PARP inhibitor sensitivity driven by RNASEH2B loss in prostate cancer.
Literature	GENERIF_SUMMARY	REST Inactivation and Coexpression of ASCL1 and POU3F4 Are Necessary for the Complete Transformation of RB1/TP53-Inactivated Lung Adenocarcinoma into Neuroendocrine Carcinoma.
Literature	GENERIF_SUMMARY	Retinoblastoma patient-derived stem cells-an in vivo model to study the role of RB1 in adipogenesis.
Literature	GENERIF_SUMMARY	Hepatocellular carcinoma evades RB1-induced senescence by activating the FOXM1-FOXO1 axis.
Literature	GENERIF_SUMMARY	pRB immunostaining in the differential diagnosis between pleomorphic xanthoastrocytoma and glioblastoma with giant cells.
Literature	GENERIF_SUMMARY	Five novel RB1 gene mutations and genotype-phenotype correlations in Chinese children with retinoblastoma.
Literature	GENERIF_SUMMARY	Novel insights into RB1 mutation.
Literature	GENERIF_SUMMARY	RB1-deficient squamous cell carcinoma
Literature	GENERIF_SUMMARY	Inactivation of RB1, CDKN2A, and TP53 have distinct effects on genomic stability at side-by-side comparison in karyotypically normal cells.
Literature	GENERIF_SUMMARY	Retinoblastoma protein as an intrinsic BRD4 inhibitor

Main Category	Category	Terms
		modulates small molecule BET inhibitor sensitivity in cancer.
Literature	GENERIF_SUMMARY	Simultaneous depletion of RB, RBL1 and RBL2 affects endoderm differentiation of human embryonic stem cells.
Literature	GENERIF_SUMMARY	Epstein-Barr virus miR-BART2-5p and miR-BART11-5p regulate cell proliferation, apoptosis, and migration by targeting RB and p21 in gastric carcinoma.
Literature	GENERIF_SUMMARY	Spontaneous Transformation of a p53 and Rb-Defective Human Fallopian Tube Epithelial Cell Line after Long Passage with Features of High-Grade Serous Carcinoma.
Literature	GENERIF_SUMMARY	Phosphohistidine signaling promotes FAK-RB1 interaction and growth factor-independent proliferation of esophageal squamous cell carcinoma.
Literature	GENERIF_SUMMARY	SETDB1 Modulates Degradation of Phosphorylated RB and Anticancer Efficacy of CDK4/6 Inhibitors.
Literature	GENERIF_SUMMARY	Onychomatricoma
Literature	GENERIF_SUMMARY	Mutant-RB1 circulating tumor DNA in the blood of unilateral retinoblastoma patients
Literature	GENERIF_SUMMARY	RB1 alteration and poor prognosis in women with high-grade neuroendocrine carcinoma of the uterine cervix
Literature	GENERIF_SUMMARY	FAK-mediated phosphorylation at Y464 regulates p85beta nuclear translocation to promote tumorigenesis of ccRCC by repressing RB1 expression.
Literature	GENERIF_SUMMARY	miR-4529-3p Promotes the Progression of Retinoblastoma by Inhibiting RB1 Expression and Activating the ERK Signaling Pathway.
Literature	GENERIF_SUMMARY	Association of Merkel Cell Polyomavirus Status With p53, RB1, and PD-L1 Expression and Patient Prognosis in Merkel Cell Carcinomas
Literature	GENERIF_SUMMARY	RB1-promoter methylation in glioblastoma
Literature	GENERIF_SUMMARY	Diagnostic Utility of TSSC3 and RB1 Immunohistochemistry in Hydatidiform Mole.
Literature	GENERIF_SUMMARY	Effects of the Overexpression of Progesterone Receptors on a Precancer p53 and Rb-Defective Human Fallopian Tube Epithelial Cell Line.
Literature	GENERIF_SUMMARY	Germline HPF1 retrogene insertion in RB1 gene involved in cancer predisposition.
Literature	GENERIF_SUMMARY	Integrative analysis of multi-omics data reveals inhibition of RB1 signaling promotes apatinib resistance of hepatocellular carcinoma.
Literature	GENERIF_SUMMARY	An overview of RB1 transcript alterations detected during retinoblastoma genetic screening.

Main Category	Category	Terms
Literature	GENERIF_SUMMARY	NEIL3 promotes cell proliferation of ccRCC via the cyclin D1-Rb-E2F1 feedback loop regulation.
Literature	GENERIF_SUMMARY	Investigation of the methylation changes in the promoter region of RB1 gene in retinoblastoma
Literature	GENERIF_SUMMARY	Analysis of E2F1 single-nucleotide polymorphisms reveals deleterious non-synonymous substitutions that disrupt E2F1-RB protein interaction in cancer.
Literature	GENERIF_SUMMARY	Novel insights into RB1 in prostate cancer lineage plasticity and drug resistance.
Literature	GENERIF_SUMMARY	P53 and Rb Aberrations in Small Cell Lung Cancer (SCLC)
Literature	GENERIF_SUMMARY	Retinoblastoma-associated protein is important for TRIM24-mediated activation of the mTOR signaling pathway through DUSP2 action in prostate cancer.
Literature	GENERIF_SUMMARY	Heterozygous RB1 mutation enhanced ATP production in human iPSC-derived retinal organoids.
Literature	GENERIF_SUMMARY	Molecular monitoring by CDKN2A/p16INK4A and RB1 gene methylation in breast cancer.
Literature	GENERIF_SUMMARY	Retinoblastoma caused by an RB1 variant with unusually low penetrance in a Danish family.
Literature	GENERIF_SUMMARY	RB1 Genetic Alterations in Estrogen Receptor-Positive Breast Carcinomas
Literature	GENERIF_SUMMARY	An intermediate Rb-E2F activity state safeguards proliferation commitment.
Literature	GENERIF_SUMMARY	The G1-S transition is promoted by Rb degradation via the E3 ligase UBR5.
Literature	GENERIF_SUMMARY	E2F target genes display elevated levels in Rb-/- MEFs. Absence of functional Rb is sufficient for S-phase entry under growth-limiting conditions, indicating that the E2F complexes containing Rb protein may be rate limiting for the G1/S transition.
Literature	GENERIF_SUMMARY	Rb interacts with the AP-2 transcription factor and activates the E-cadherin gene expression
Literature	GENERIF_SUMMARY	Rb is a coactivator of AP-2 in epithelial cells
Pathways	KEGG_PATHWAY	Endocrine resistance
Pathways	KEGG_PATHWAY	Cell cycle
Pathways	KEGG_PATHWAY	Cellular senescence
Pathways	KEGG_PATHWAY	Cushing syndrome
Pathways	KEGG_PATHWAY	Hepatitis C
Pathways	KEGG_PATHWAY	Hepatitis B
Pathways	KEGG_PATHWAY	Human cytomegalovirus infection

Main Category	Category	Terms
Pathways	KEGG_PATHWAY	Human papillomavirus infection
Pathways	KEGG_PATHWAY	Human T-cell leukemia virus 1 infection
Pathways	KEGG_PATHWAY	Kaposi sarcoma-associated herpesvirus infection
Pathways	KEGG_PATHWAY	Epstein-Barr virus infection
Pathways	KEGG_PATHWAY	Pathways in cancer
Pathways	KEGG_PATHWAY	Viral carcinogenesis
Pathways	KEGG_PATHWAY	Chemical carcinogenesis - receptor activation
Pathways	KEGG_PATHWAY	Pancreatic cancer
Pathways	KEGG_PATHWAY	Glioma
Pathways	KEGG_PATHWAY	Prostate cancer
Pathways	KEGG_PATHWAY	Melanoma
Pathways	KEGG_PATHWAY	Bladder cancer
Pathways	KEGG_PATHWAY	Chronic myeloid leukemia
Pathways	KEGG_PATHWAY	Small cell lung cancer
Pathways	KEGG_PATHWAY	Non-small cell lung cancer
Pathways	KEGG_PATHWAY	Breast cancer
Pathways	KEGG_PATHWAY	Hepatocellular carcinoma
Pathways	KEGG_PATHWAY	Gastric cancer
Pathways	REACTOME_PATHWAY	Inhibition of replication initiation of damaged DNA by RB1/E2F1
Pathways	REACTOME_PATHWAY	E2F mediated regulation of DNA replication
Pathways	REACTOME_PATHWAY	Cell-Cell communication
Pathways	REACTOME_PATHWAY	Cell Cycle
Pathways	REACTOME_PATHWAY	Disease
Pathways	REACTOME_PATHWAY	APC/C-mediated degradation of cell cycle proteins
Pathways	REACTOME_PATHWAY	APC/C:Cdh1 mediated degradation of Cdc20 and other APC/C:Cdh1 targeted proteins in late mitosis/early G1
Pathways	REACTOME_PATHWAY	Epigenetic regulation of gene expression
Pathways	REACTOME_PATHWAY	Generic Transcription Pathway
Pathways	REACTOME_PATHWAY	Cellular responses to stress
Pathways	REACTOME_PATHWAY	Condensation of Prophase Chromosomes
Pathways	REACTOME_PATHWAY	Cellular Senescence
Pathways	REACTOME_PATHWAY	Formation of Senescence-Associated Heterochromatin Foci (SAHF)

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	Oncogene Induced Senescence
Pathways	REACTOME_PATHWAY	DNA Damage/Telomere Stress Induced Senescence
Pathways	REACTOME_PATHWAY	Adherens junctions interactions
Pathways	REACTOME_PATHWAY	Cell-cell junction organization
Pathways	REACTOME_PATHWAY	Cell junction organization
Pathways	REACTOME_PATHWAY	Regulation of mitotic cell cycle
Pathways	REACTOME_PATHWAY	Mitotic G1 phase and G1/S transition
Pathways	REACTOME_PATHWAY	Diseases of signal transduction by growth factor receptors and second messengers
Pathways	REACTOME_PATHWAY	Infectious disease
Pathways	REACTOME_PATHWAY	Mitotic Prophase
Pathways	REACTOME_PATHWAY	M Phase
Pathways	REACTOME_PATHWAY	Phosphorylation of proteins involved in G1/S transition by active Cyclin E:Cdk2 complexes
Pathways	REACTOME_PATHWAY	Cyclin E associated events during G1/S transition
Pathways	REACTOME_PATHWAY	G1/S Transition
Pathways	REACTOME_PATHWAY	Cyclin D associated events in G1
Pathways	REACTOME_PATHWAY	G1 Phase
Pathways	REACTOME_PATHWAY	S Phase
Pathways	REACTOME_PATHWAY	Cell Cycle, Mitotic
Pathways	REACTOME_PATHWAY	Cyclin A:Cdk2-associated events at S phase entry
Pathways	REACTOME_PATHWAY	RNA Polymerase II Transcription
Pathways	REACTOME_PATHWAY	Gene expression (Transcription)
Pathways	REACTOME_PATHWAY	Transcriptional regulation by RUNX2
Pathways	REACTOME_PATHWAY	RUNX2 regulates osteoblast differentiation
Pathways	REACTOME_PATHWAY	RUNX2 regulates bone development
Pathways	REACTOME_PATHWAY	Cellular responses to stimuli
Pathways	REACTOME_PATHWAY	Aberrant regulation of mitotic G1/S transition in cancer due to RB1 defects
Pathways	REACTOME_PATHWAY	Defective binding of RB1 mutants to E2F1,(E2F2, E2F3)
Pathways	REACTOME_PATHWAY	Defective translocation of RB1 mutants to the nucleus
Pathways	REACTOME_PATHWAY	Diseases of mitotic cell cycle
Pathways	REACTOME_PATHWAY	SARS-CoV-1 Infection
Pathways	REACTOME_PATHWAY	SARS-CoV Infections

Main Category	Category	Terms
Pathways	REACTOME_PATHWAY	SARS-CoV-1 Genome Replication and Transcription
Pathways	REACTOME_PATHWAY	Replication of the SARS-CoV-1 genome
Pathways	REACTOME_PATHWAY	Aberrant regulation of mitotic exit in cancer due to RB1 defects
Pathways	REACTOME_PATHWAY	Aberrant regulation of mitotic cell cycle due to RB1 defects
Pathways	REACTOME_PATHWAY	SARS-CoV-2 Infection
Pathways	REACTOME_PATHWAY	SARS-CoV-2 Genome Replication and Transcription
Pathways	REACTOME_PATHWAY	Replication of the SARS-CoV-2 genome
Pathways	REACTOME_PATHWAY	Signaling by ALK in cancer
Pathways	REACTOME_PATHWAY	Signaling by ALK fusions and activated point mutants
Pathways	REACTOME_PATHWAY	Nuclear events stimulated by ALK signaling in cancer
Pathways	REACTOME_PATHWAY	Regulation of Homotypic Cell-Cell Adhesion
Pathways	REACTOME_PATHWAY	Regulation of CDH1 Expression and Function
Pathways	REACTOME_PATHWAY	Regulation of Expression and Function of Type I Classical Cadherins
Pathways	REACTOME_PATHWAY	Regulation of CDH1 Gene Transcription
Pathways	REACTOME_PATHWAY	Positive Regulation of CDH1 Gene Transcription
Pathways	REACTOME_PATHWAY	Early SARS-CoV-2 Infection Events
Pathways	REACTOME_PATHWAY	Epigenetic regulation of gene expression by MLL3 and MLL4 complexes
Pathways	REACTOME_PATHWAY	Viral Infection Pathways
Pathways	REACTOME_PATHWAY	MLL4 and MLL3 complexes regulate expression of PPARG target genes in adipogenesis and hepatic steatosis
Pathways	REACTOME_PATHWAY	Epigenetic regulation of adipogenesis genes by MLL3 and MLL4 complexes
Pathways	REACTOME_PATHWAY	Epigenetic regulation by WDR5-containing histone modifying complexes
Pathways	WIKIPATHWAYS	Androgen receptor signaling
Pathways	WIKIPATHWAYS	miRNA regulation of DNA damage response
Pathways	WIKIPATHWAYS	Cell cycle
Pathways	WIKIPATHWAYS	Integrated cancer pathway
Pathways	WIKIPATHWAYS	Integrated breast cancer pathway
Pathways	WIKIPATHWAYS	miR targeted genes in lymphocytes
Pathways	WIKIPATHWAYS	miR targeted genes in muscle cell

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Thyroid stimulating hormone TSH signaling
Pathways	WIKIPATHWAYS	Glioblastoma signaling
Pathways	WIKIPATHWAYS	Androgen receptor network in prostate cancer
Pathways	WIKIPATHWAYS	Adipogenesis
Pathways	WIKIPATHWAYS	Spinal cord injury
Pathways	WIKIPATHWAYS	Retinoblastoma gene in cancer
Pathways	WIKIPATHWAYS	Aryl hydrocarbon receptor pathway
Pathways	WIKIPATHWAYS	Bladder cancer
Pathways	WIKIPATHWAYS	Fas ligand pathway and stress induction of heat shock proteins
Pathways	WIKIPATHWAYS	Photodynamic therapy induced AP 1 survival signaling
Pathways	WIKIPATHWAYS	H19 Rb E2F1 and CDK beta catenin in colorectal cancer
Pathways	WIKIPATHWAYS	Prader Willi and Angelman syndrome
Pathways	WIKIPATHWAYS	Tumor suppressor activity of SMARCB1
Pathways	WIKIPATHWAYS	Non small cell lung cancer
Pathways	WIKIPATHWAYS	Breast cancer pathway
Pathways	WIKIPATHWAYS	Pancreatic adenocarcinoma pathway
Pathways	WIKIPATHWAYS	Effect of progerin on genes involved in progeria
Pathways	WIKIPATHWAYS	Ciliary landscape
Pathways	WIKIPATHWAYS	G1 to S cell cycle control
Pathways	WIKIPATHWAYS	Small cell lung cancer
Pathways	WIKIPATHWAYS	Head and neck squamous cell carcinoma
Pathways	WIKIPATHWAYS	Melanoma
Pathways	WIKIPATHWAYS	Overlap between signal transduction pathways contributing to LMNA laminopathies
Pathways	WIKIPATHWAYS	7q11 23 copy number variation syndrome
Pathways	WIKIPATHWAYS	Glycolysis in senescence
Pathways	WIKIPATHWAYS	Sphingolipid metabolism in senescence
Pathways	WIKIPATHWAYS	Prostaglandin and leukotriene metabolism in senescence
Pathways	WIKIPATHWAYS	ID signaling
Pathways	WIKIPATHWAYS	TROP2 regulatory signaling
Pathways	WIKIPATHWAYS	8q11 23 RB1CC1 copy number variation
Pathways	WIKIPATHWAYS	Omega 6 fatty acids in senescence

Main Category	Category	Terms
Pathways	WIKIPATHWAYS	Cancer pathways
Pathways	WIKIPATHWAYS	Senescence and autophagy in cancer
Pathways	WIKIPATHWAYS	DNA damage response
Protein Domains	INTERPRO	RB_B
Protein Domains	INTERPRO	RB_A
Protein Domains	INTERPRO	Cyclin-like_dom
Protein Domains	INTERPRO	RB_C
Protein Domains	INTERPRO	RB_N
Protein Domains	INTERPRO	RB_fam
Protein Domains	INTERPRO	Cyclin-like_sf
Protein Domains	SMART	CYCLIN
Protein Domains	SMART	DUF3452
Protein Domains	SMART	RB_A
Protein Domains	SMART	Rb_C